



BRIDGING THE GAP BETWEEN ENTERPRISE AI AND GENERATIVE AI

An Analysis of Time series Foundational models for forecasting highly volatile Cryptocurrency Close prices.

Capstone Project By

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Agenda

- Enterprise AI VS Generative AI
- Introduction to Time Series Foundation Models
- Project Objective
- Outlines – Model and Evaluation Metrics
- Exploratory Data Analysis
- Experiment Results
- Evaluation Metrics Table
- A Note on scaled vs unscaled inputs
- Curious Case of Covariates
- Potential Solution for Covariates Inclusion
- Our Learning
- Future Work
- Appendix

Enterprise AI VS Generative AI – Use Cases POV

Enterprise AI

- **Classification:** Spam detection, disease diagnosis, credit scoring
- **Regression:** Sales forecasting, price prediction
- **Rule-based Systems:** Chatbots, recommendation engines (based on heuristics)
- **Clustering:** Customer segmentation, fraud pattern detection
- **Optimization:** Route planning, inventory management

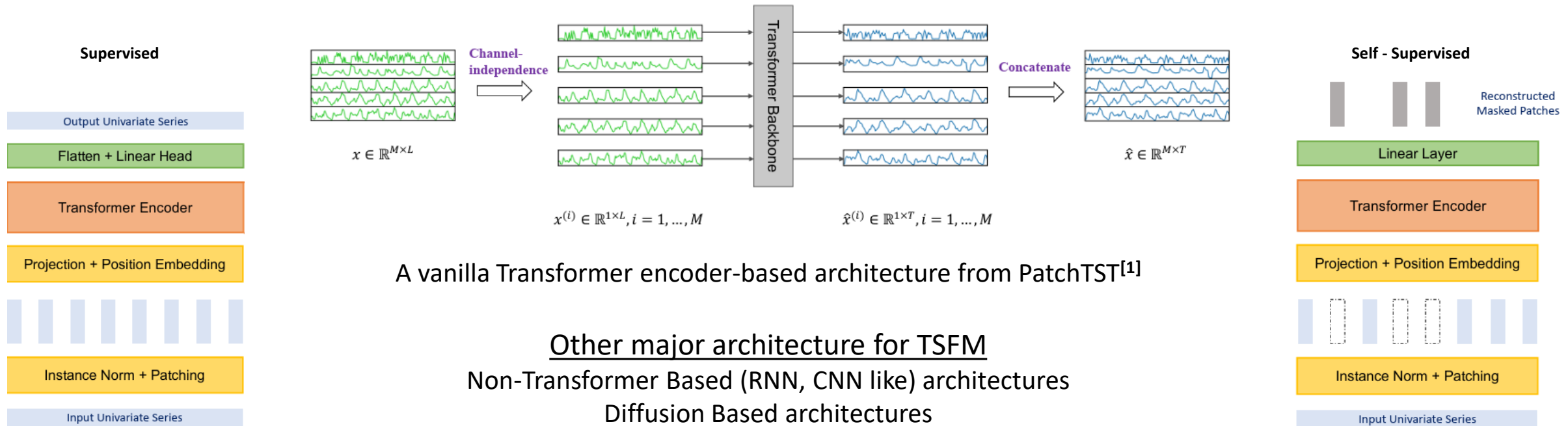
Generative AI

- **Text Generation:** ChatGPT, legal drafting, content creation
- **Image Generation:** DALL·E, Midjourney, product mockups
- **Audio & Music:** AI voice synthesis, background music generation
- **Code Generation:** GitHub Copilot, data pipeline automation
- **Synthetic Data:** Augmenting training data for healthcare or rare events

Till date, Enterprise AI accounts for over **95% of decision-making** in addressing real-world challenges faced by organizations wherever AI/ML is applicable

Introduction to Time Series Foundation Models (TSFMs) – I

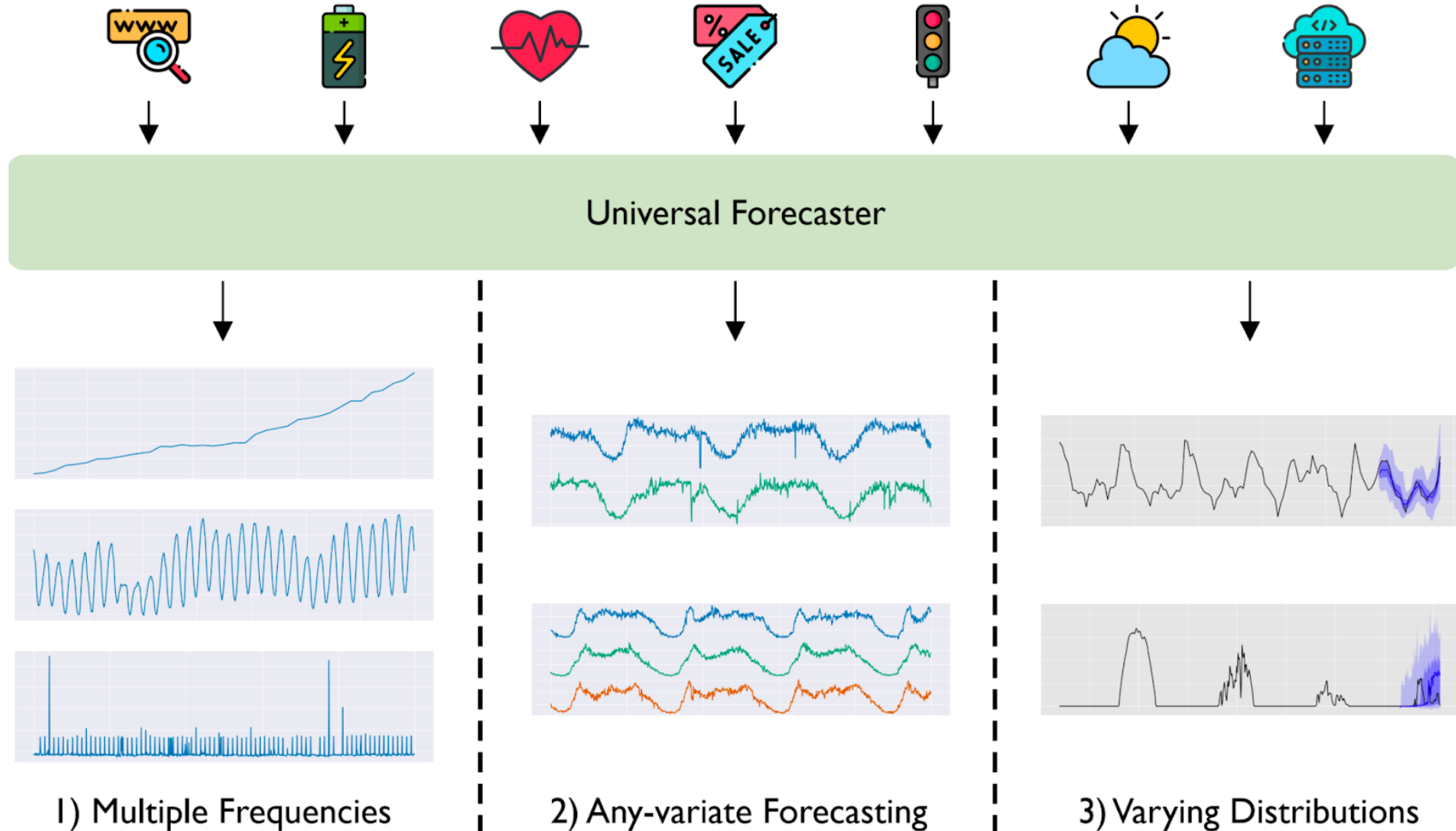
- TSFMs are pre-trained models designed to generalize across multiple tasks and domains using large-scale time series data
- Inspired by the success of LLMs and vision models
- Adopt similar strategies like pre-training, fine-tuning, prompt engineering for time series problems
- Support diverse applications – forecasting, classification, anomaly detection
- Applicable across sectors like finance, healthcare, traffic, and climate



TSFMs bring the foundation model revolution to time series with the goal of general-purpose, scalable, and domain-adaptable intelligence.

Introduction to Time Series Foundation Models (TSFMs) - II

The Ultimate Goal of Foundation Models^[7]



Project Objective

Problem Statement

Evaluation of Time Series Foundation Models for Daily Cryptocurrency Price Forecasting

Data

- Daily Close price (in USD) of 4 cryptocurrencies with varied volume, initiation time and popularity
- Sourced from Yahoo Finance API (YFinance)

Bitcoin (BTC)

2014 - 2024

Binance (BND)

2017 - 2024

Ethereum (ETH)

2017 - 2024

Solana (SOL)

2020 - 2024

Models

- Three popular and well backed Time Series Foundation Models

TimesFM^[2]



Chronos^[3]



Tiny Time Mixers^[4]



Moirai^[7]



Experimental Setup

- Training Period: Initiation – 2023
- Context Length: 512
- Benchmarking Models: ARIMA, Prophet, LSTM

Testing Period: 2024 (366 days)

Forecasting Horizons: 30 days*, 7 days, 1 day

*- 30 days horizon was used for Benchmarking models

Evaluation Metrics^[5]

MAPE

ME

MAE

MPE

RMSE

R-squared

Scalar Product

Return Score

Long Return

Short Return

Mean Directional Accuracy

Mean Directional Accuracy Positive

Mean Directional Accuracy Negative

Model Outline - I

TimesFM ^[2]



- A **decoder-only Transformer** model designed for universal time series forecasting
- Pre-trained on a massive and diverse time series corpus across domains
- Excels in zero-shot and few-shot forecasting without task-specific fine-tuning
- Stacked transformer layers (self-attention and feedforward layers) as the main building blocks like LLMs
- A group of contiguous time-points called as a **Patch** is treated as a token
- The goal is to forecast the $(i+1)^{\text{th}}$ patch of time-points given the i^{th} output at the end of the stacked transformer layers

Chronos ^[3]



- Modular foundation model family with **encoder and decoder Transformer** model based on the T5 (Text-to-Text Transfer Transformer) architecture to forecast future patterns
- Trained on Chronos-Data, a massive, domain-diverse dataset with over 100-billion-time steps from 1000+ datasets
- Ideal for plug-and-play multivariate forecasting for healthcare, operations, climate, and time series competitions
- The T5 family of models is a series of language models developed by Google, Chronos has adapted the model to predict unbounded set of continuous values instead of a finite dictionary of words

Using these models for cryptocurrency price prediction enables flexible, scalable, and efficient forecasting across market conditions as zero shot learning with minimal task-specific tuning

Model Outline - II

Tiny Time Mixers^[4]



- A lightweight **MLP-based models** using temporal mixing blocks and adaptive patching (**no attention mechanism**)
- TTM is pretrained on 244M samples of the Monash dataset, using 6 A100 GPUs in less than 8 hours
- Ideal for on-device forecasting, real-time applications, and edge scenarios with limited compute resources
- Inherent capability for multivariate forecasting – extra channels, exogenous variables, and known future inputs
- Novel adaptive patching helps the model generalize better across diverse data as each time series performs optimally at a specific patch length. It also uses an extra embedding layer to encode time-series frequency

Moirai^[6]



- Moirai is designed to achieve universal forecasting with **masked encoder-based time series transformers**
- It is pretrained on LOTSA – a large-scale and diverse time series dataset, comprising 27 billion observations spanning nine distinct domains
- Incorporates multiple patch size projection layers to capture temporal patterns across various frequencies
- Evaluated in both in-distribution and out-of-distribution settings, Moirai shines as a zero-shot forecaster, delivering competitive or even superior performance compared to full-shot models
- Extends the architecture to the multivariate time series by flattening– considering all variates as a single sequence

Powerful and lightweight frameworks to capture the intricacies of time series with inherent support for multivariate analysis*

* only through core development package

Evaluation Metric Outline – ML Based

Metric

Definition and Intuition

MAPE

- **Mean Absolute Percentage Error** measures the average absolute percentage difference between predicted and actual values.
- Provides the sense of forecast accuracy as a percentage error, making it easy to interpret across scales.

ME

- **Mean Error** is the average of the differences between predicted and actual values.
- Indicates whether predictions are systematically over or under the actual values (bias) but doesn't reflect error magnitude.

MAE

- **Mean Absolute Error** is the average of the absolute differences between predicted and actual values:
- Gives a clear, linear measure of prediction accuracy without amplifying large errors

MPE

- **Mean Percentage Error** is the average of the percentage differences between predicted and actual values.
- Shows the bias in predictions as a percentage, indicating whether predictions tend to overestimate or underestimate.

RMSE

- **Root Mean Squared Error** is the square root of the average of squared differences between predicted and actual values.
- Penalizes larger errors more heavily, making it sensitive to outliers and useful when big mistakes matter.

R-Squared

- **R²** measures the proportion of variance in the actual values explained by the model.
- Indicates how well the predictions fit the data, with 1 being perfect and 0 meaning no explanatory power.

Evaluation Metric Outline – Finance Based

Metric	Definition and Intuition
Scalar Product	• Measures similarity between predicted and actual return vectors • Intuition: Higher value = more alignment in direction and magnitude of price movements – Higher is better
Return Score	• Custom accuracy metric based on predicted vs actual return signs • Intuition: Reflects how often the model captures profitable trends – closer to 1 is better
Long Return	• Average return when model predicts an upward (long) move • Intuition: Shows model's effectiveness in capturing gains when it signals buy – Higher is better
Short Return	• Average return when model predicts a downward (short) move. • Intuition: Indicates how well the model performs when signaling sell – Higher is better
Mean Directional Accuracy	• Measures how often the model gets the direction of price change right • Intuition: Like a trend hit rate – not about exact price, just the up/down movement – Closer to 1 is better
Mean Directional Accuracy Positive	• MDA computed only on positive returns (when actual market moved up) • Intuition: Measures model's precision in catching upward trends – Closer to 1 is better
Mean Directional Accuracy Negative	• MDA computed only on negative returns (when actual market fell) • Intuition: Reflects how reliably the model catches downward trends – Closer to 1 is better

Exploratory Data Analysis – Close Price and Volume Traded

BTC



ETH



BNB



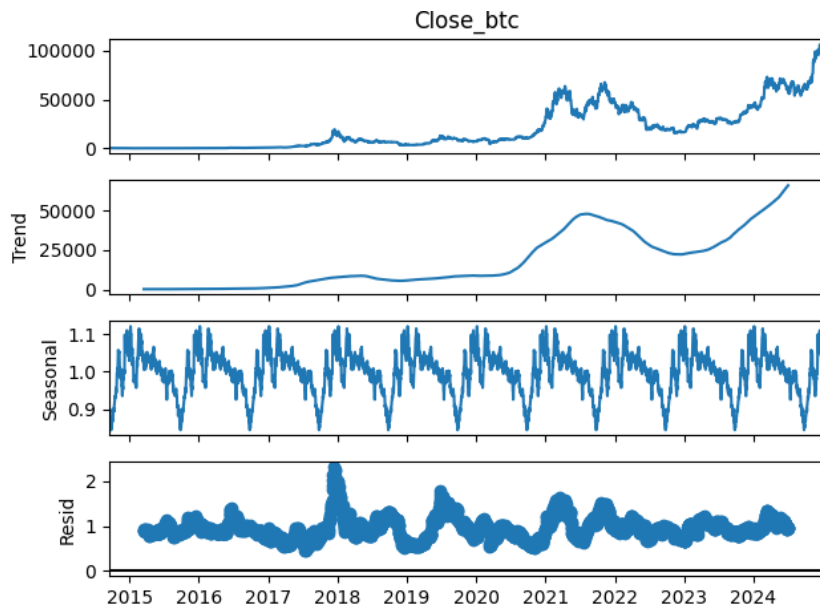
SOL



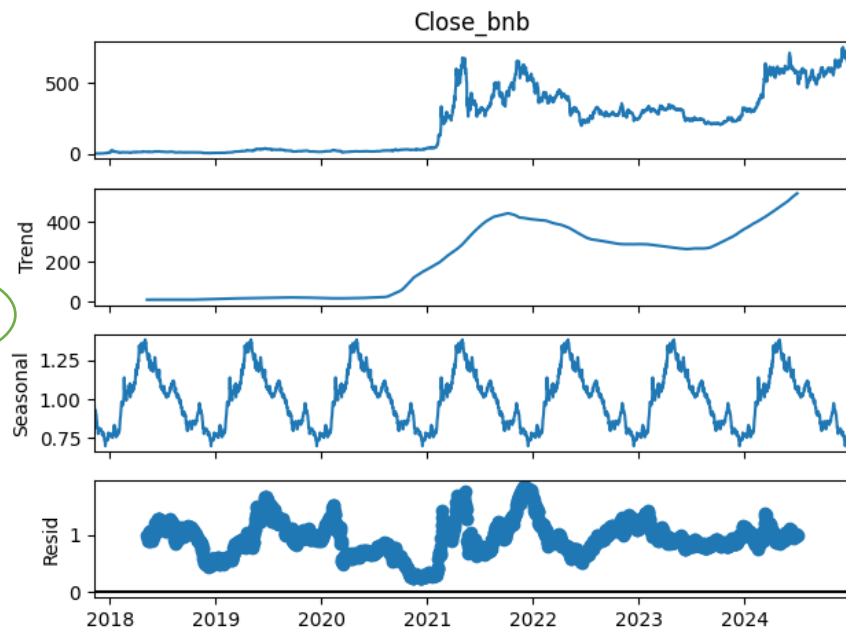
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Exploratory Data Analysis – Seasonal Decomposition

BTC

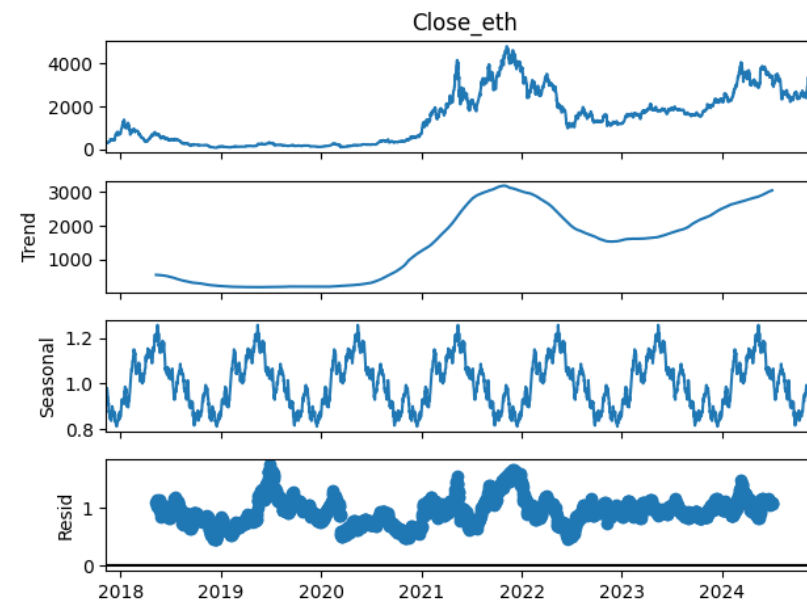


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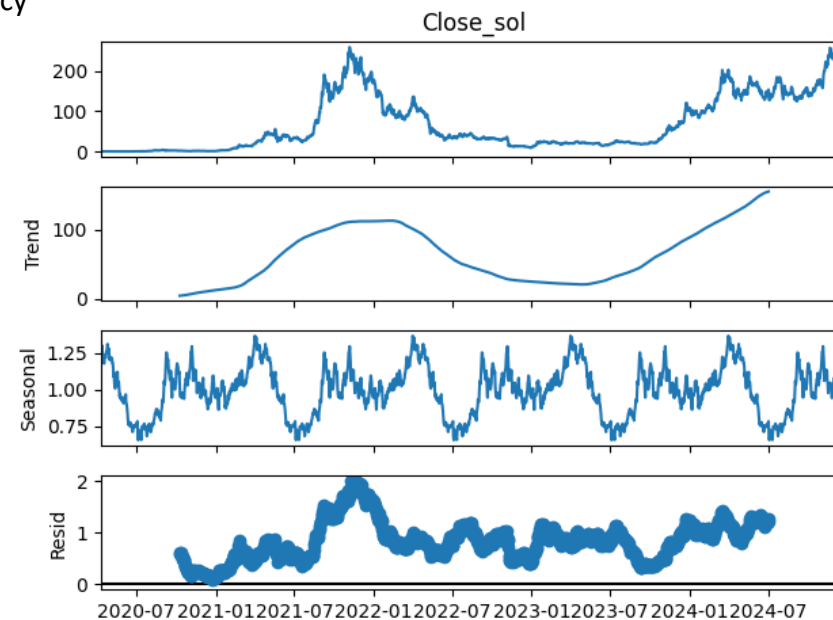


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ETH

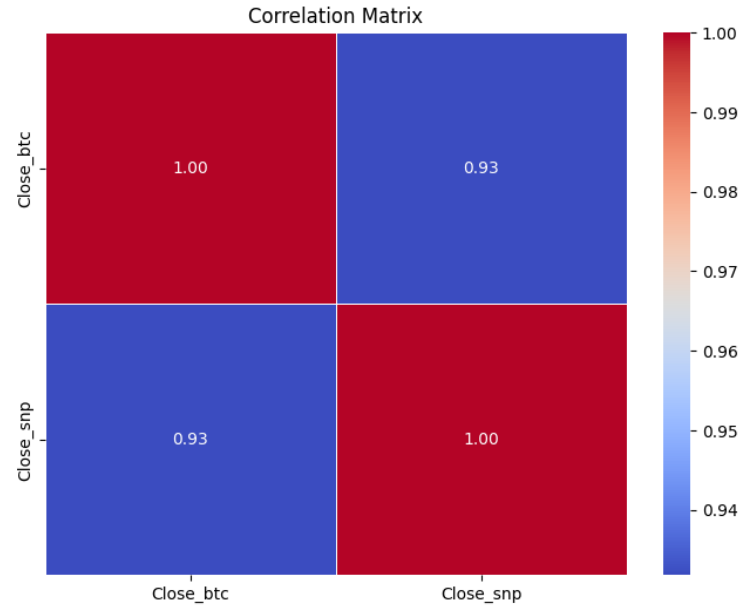


SOL

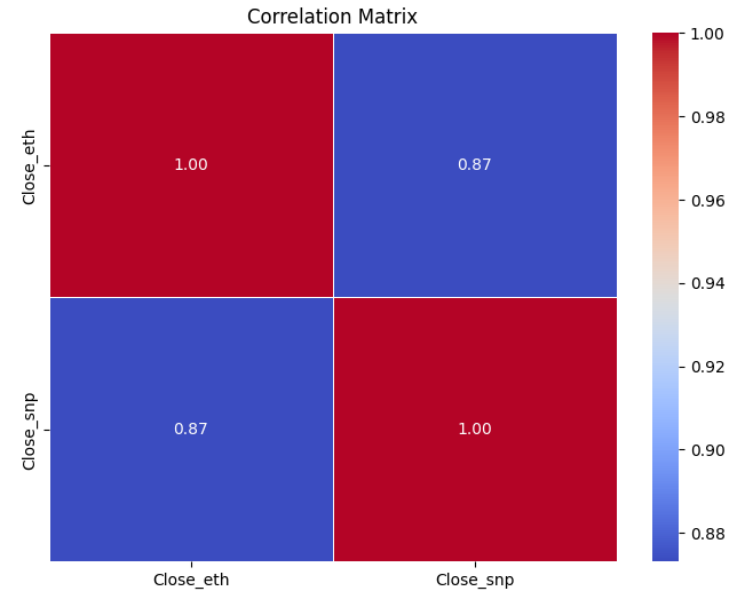


Exploratory Data Analysis – Correlation with S&P 500 Index

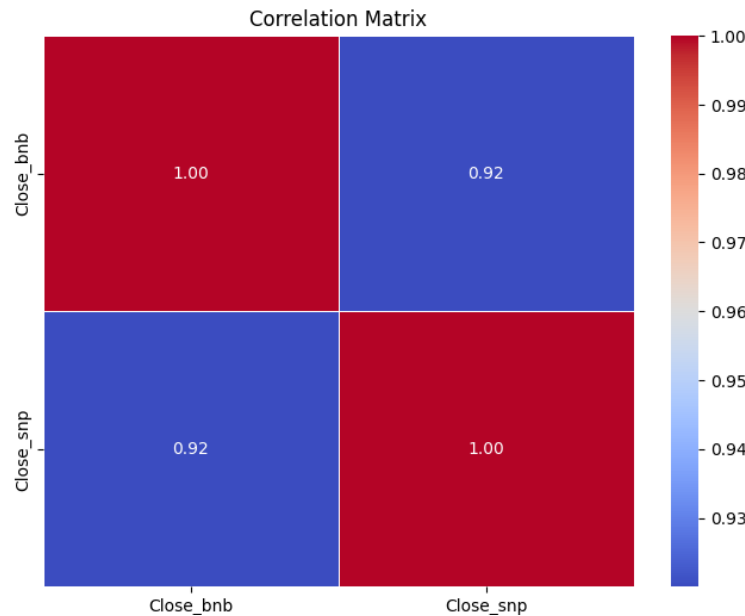
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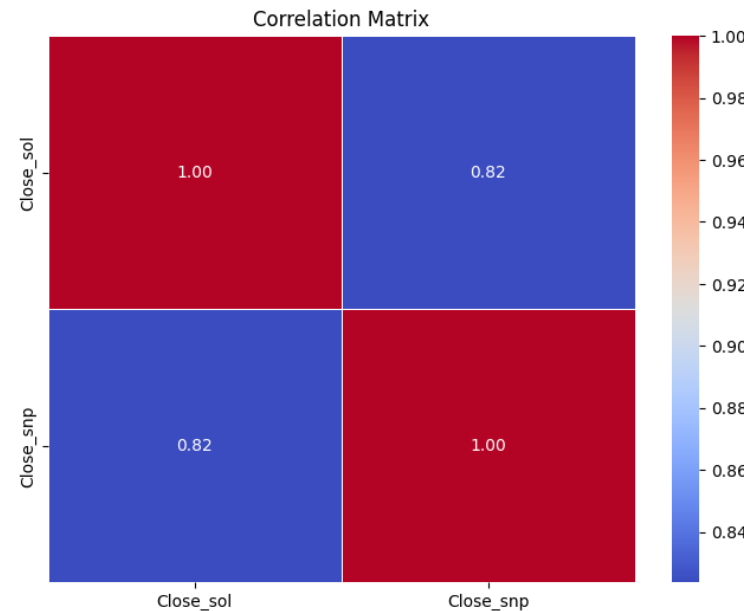
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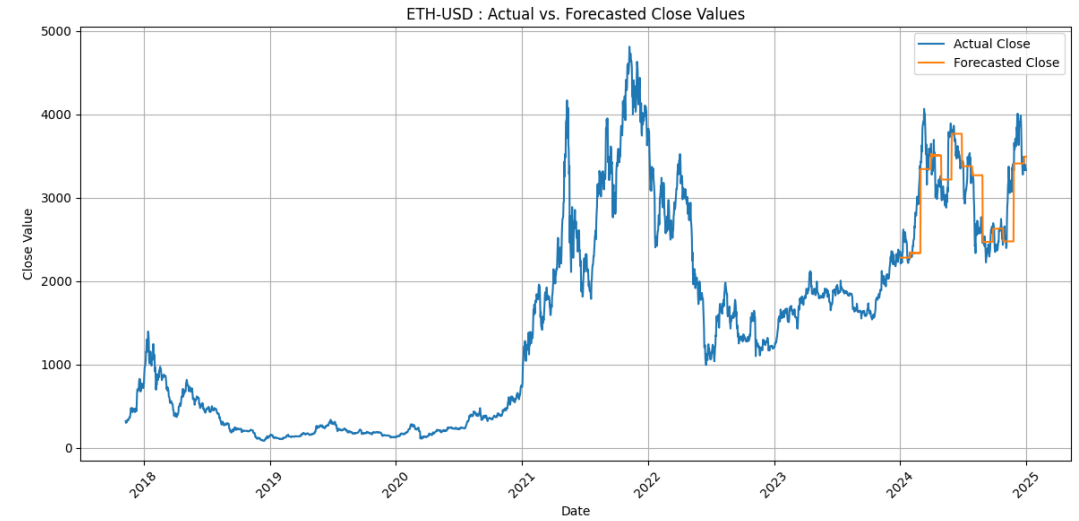
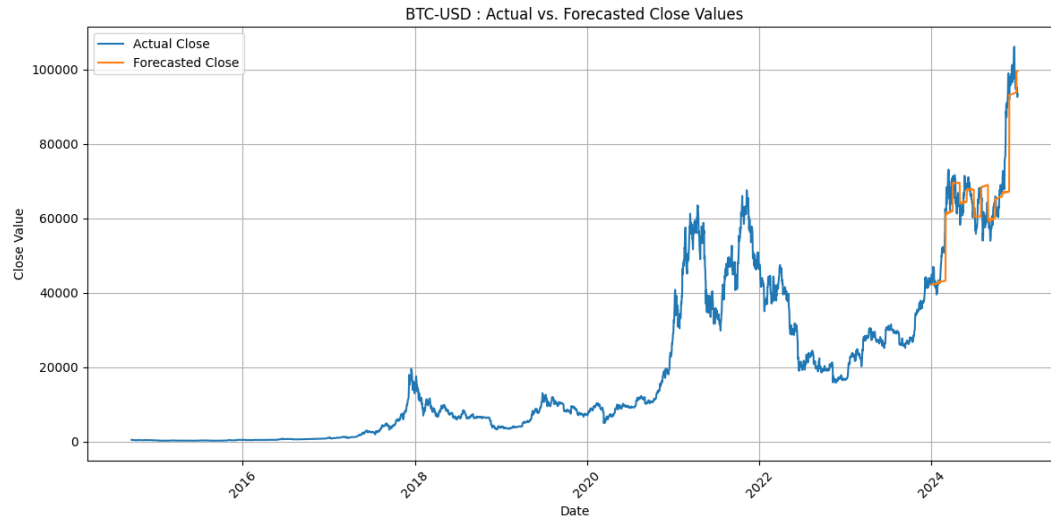


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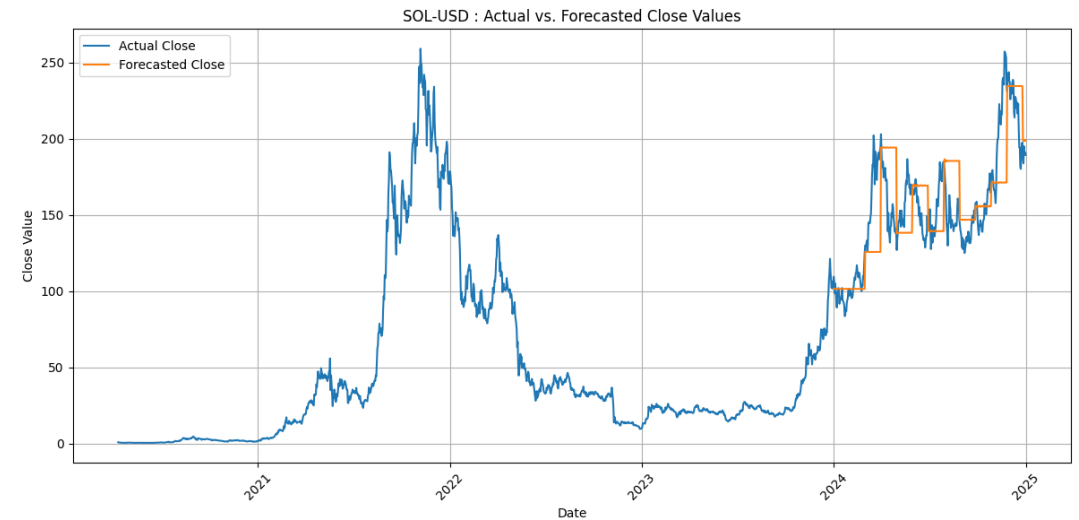
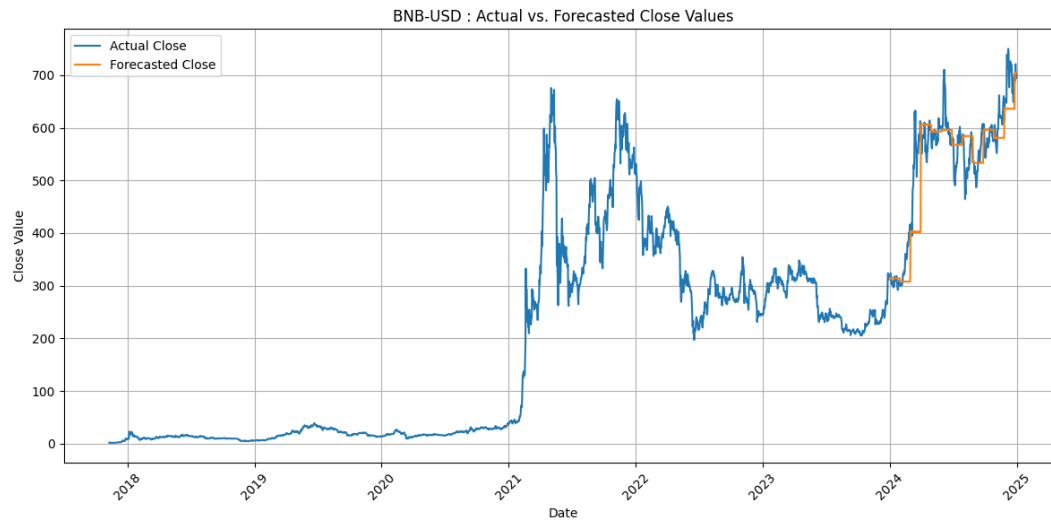


Experiment Results – Benchmark Model - ARIMA

Unscaled | Model: AutoARIMA | Horizon: 30

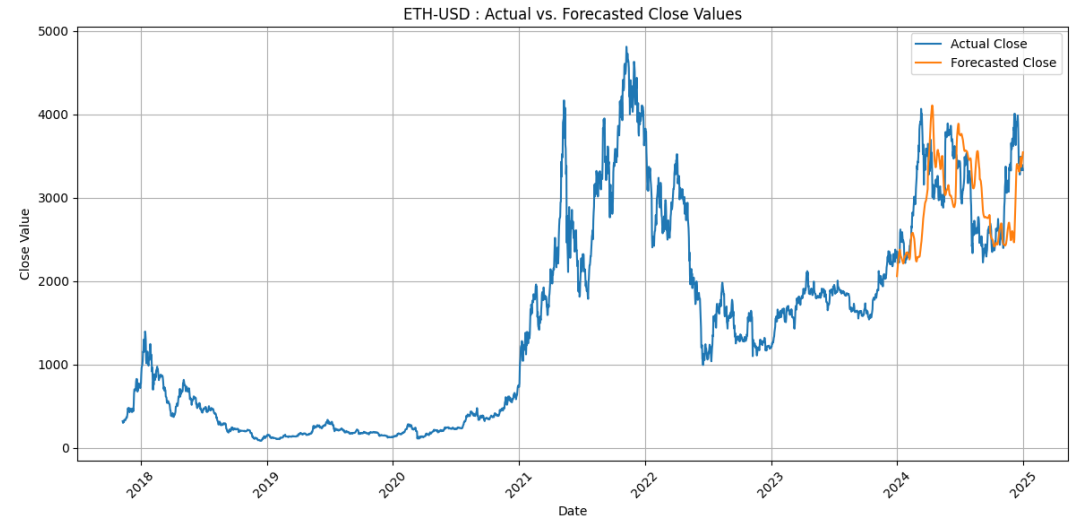
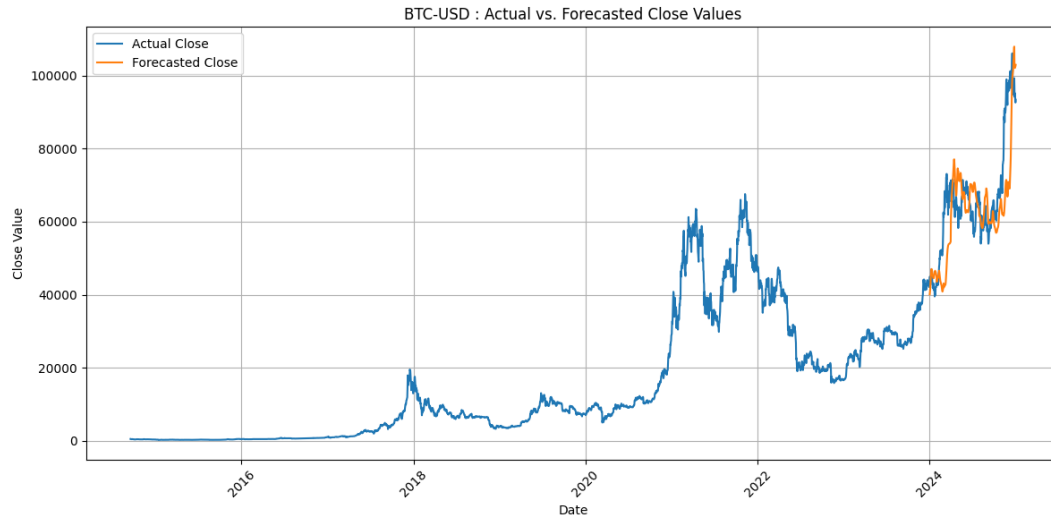


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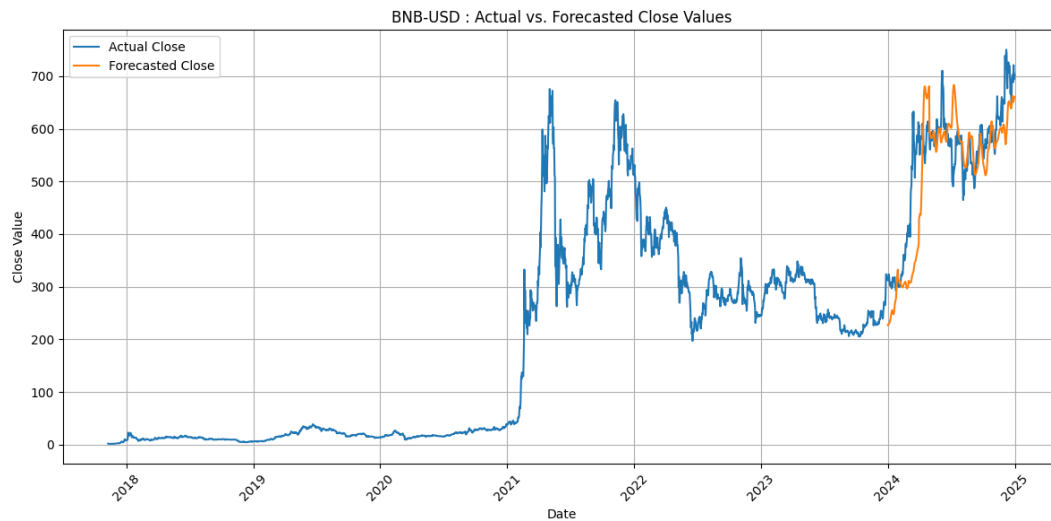


Experiment Results – Benchmark Model - LSTM

Unscaled | Model: NeuralForecastLSTM | Horizon: 30

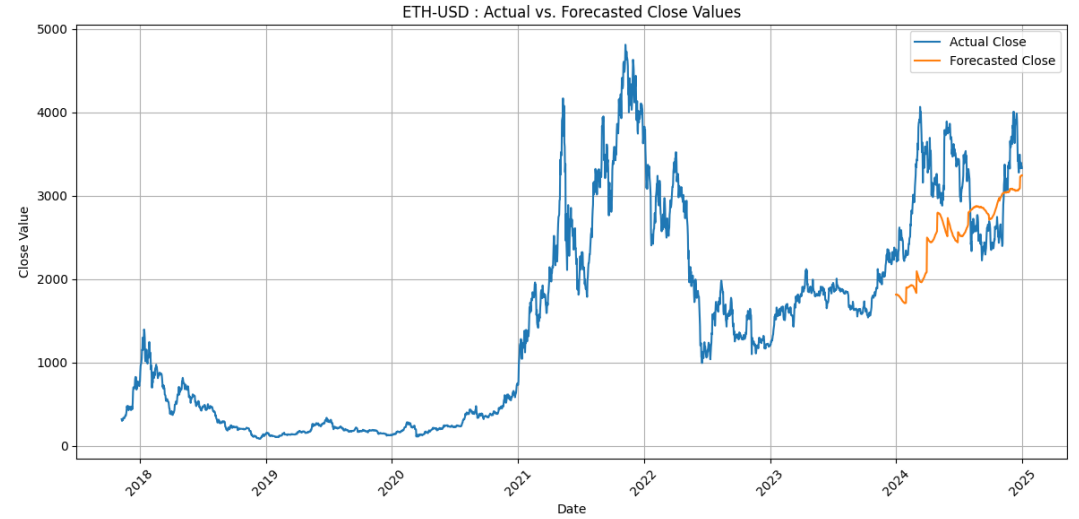
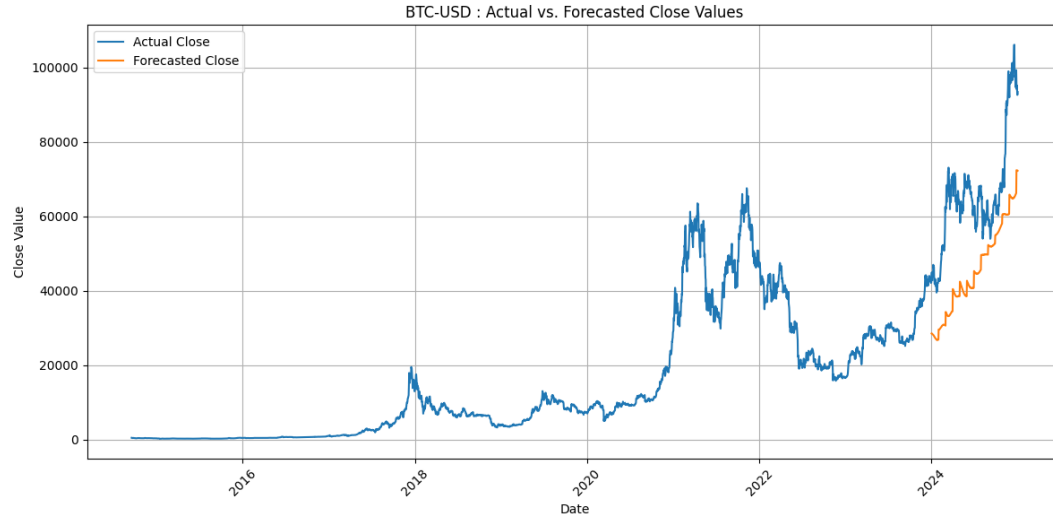


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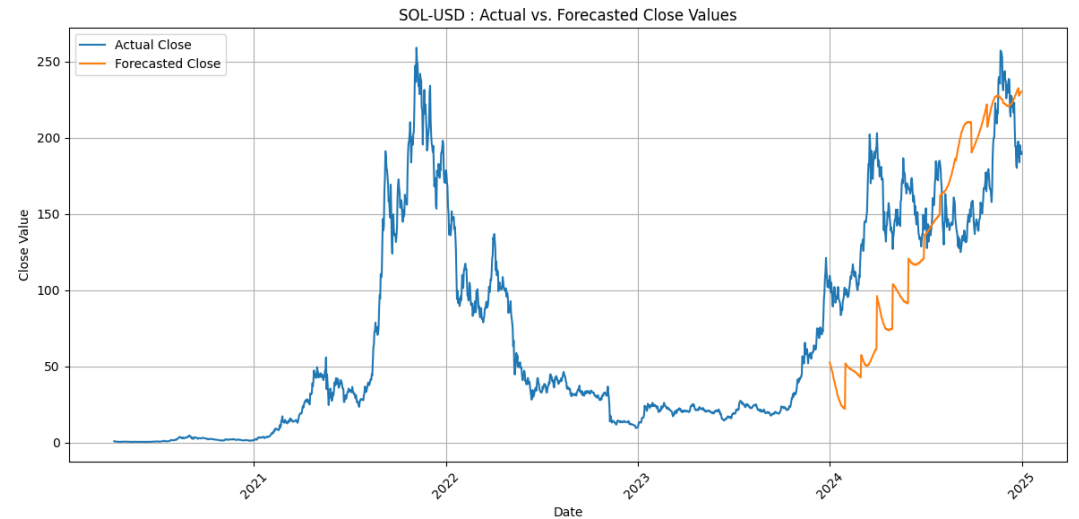
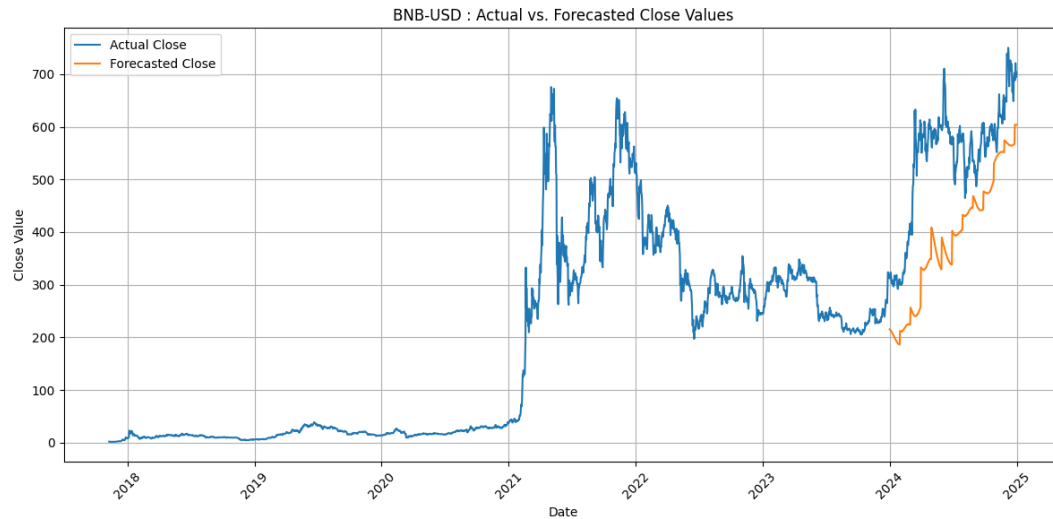


Experiment Results – Benchmark Model - Prophet

Unscaled | Model: Prophet | Horizon: 30

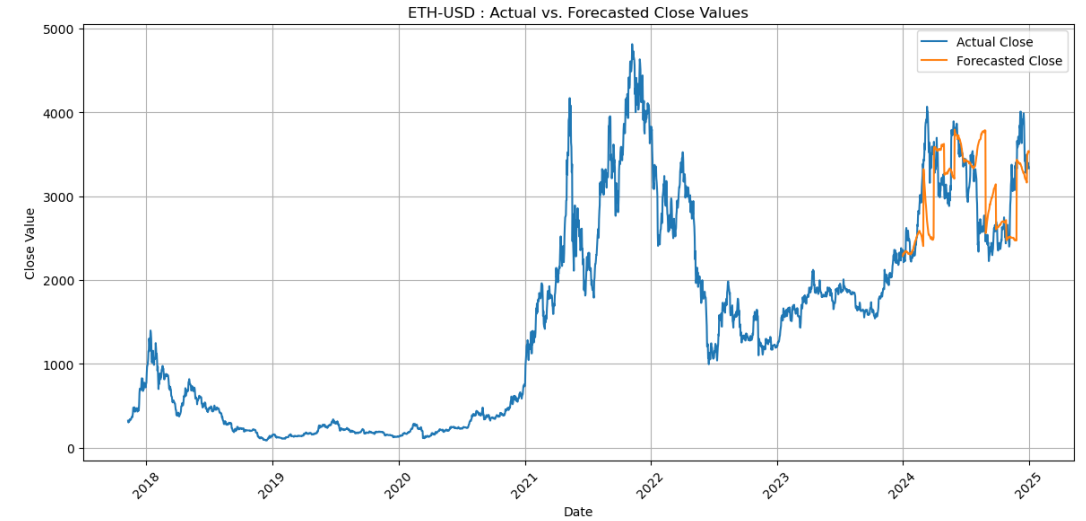
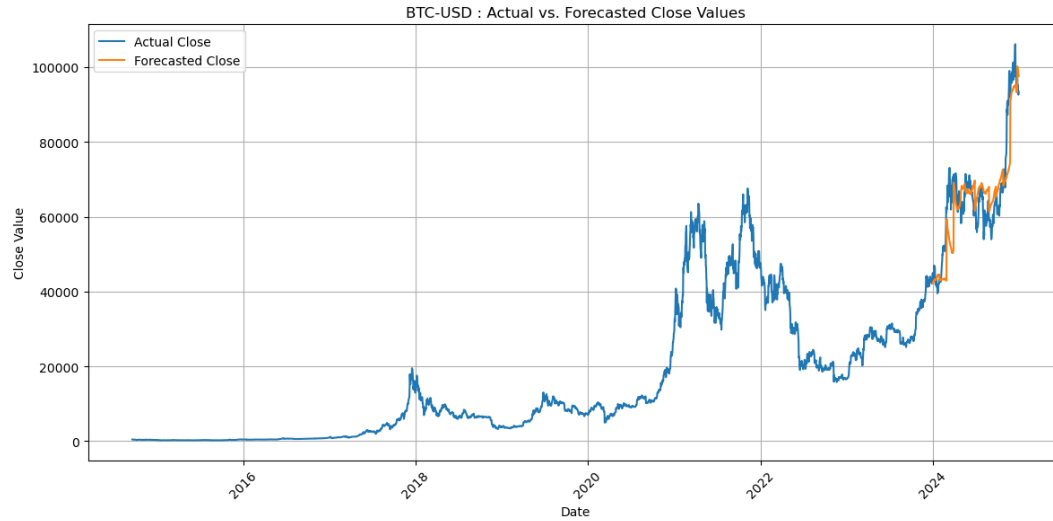


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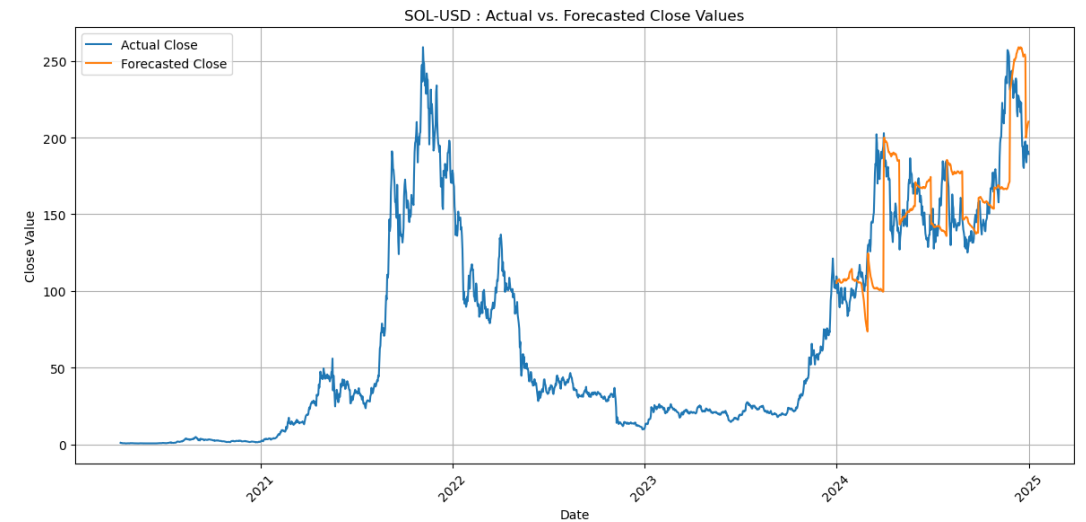
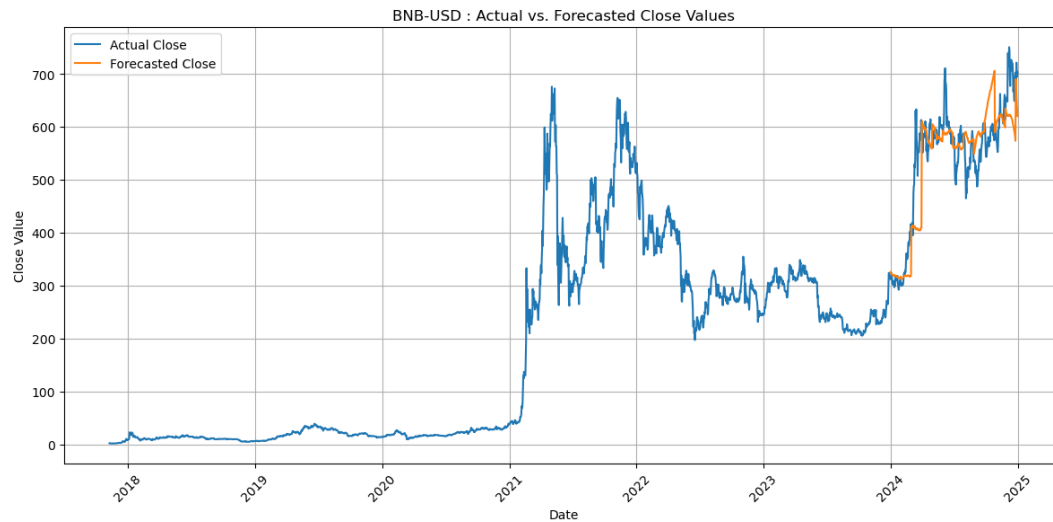


Experiment Results – TimesFM - I

Unscaled | Model: TimesFM | Horizon: 30 | Context Length: 512

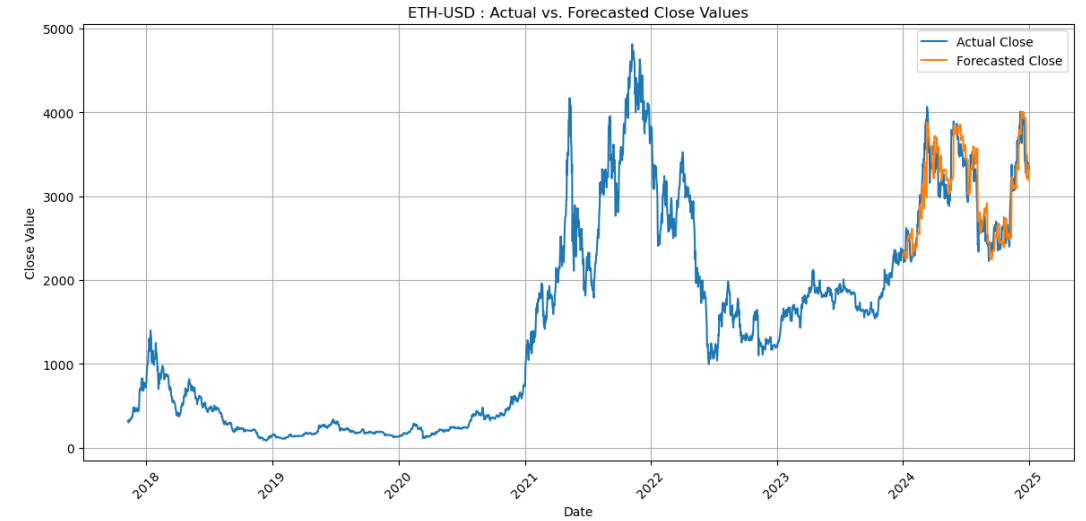
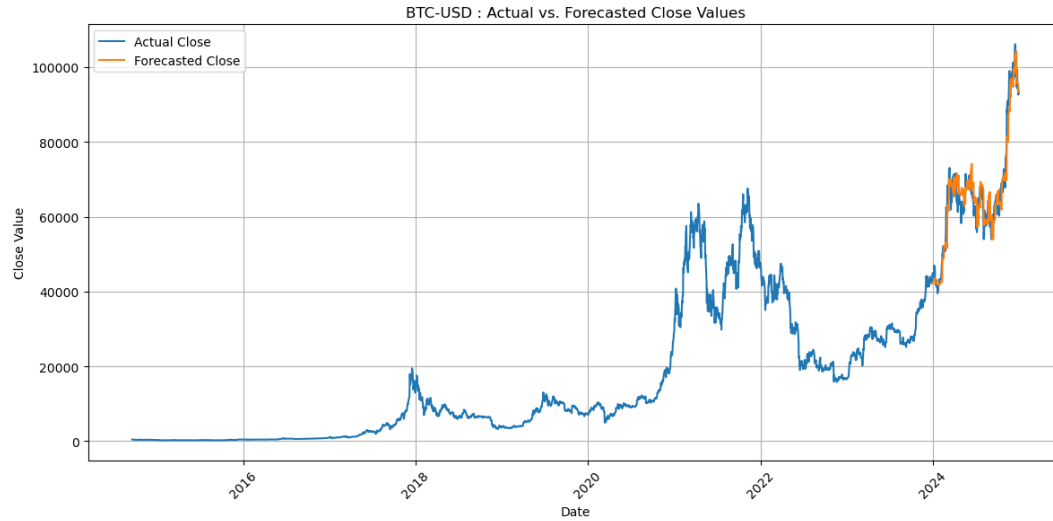


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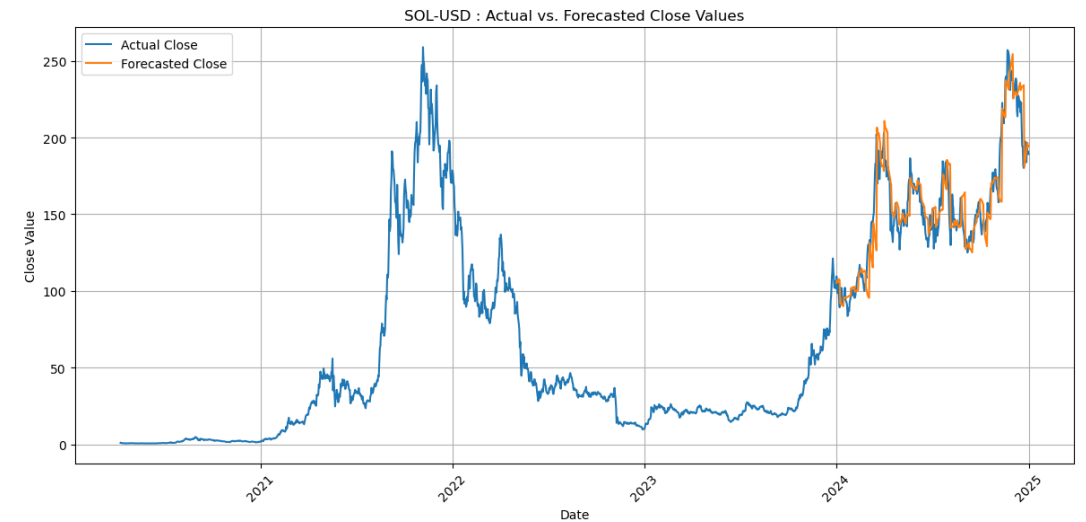
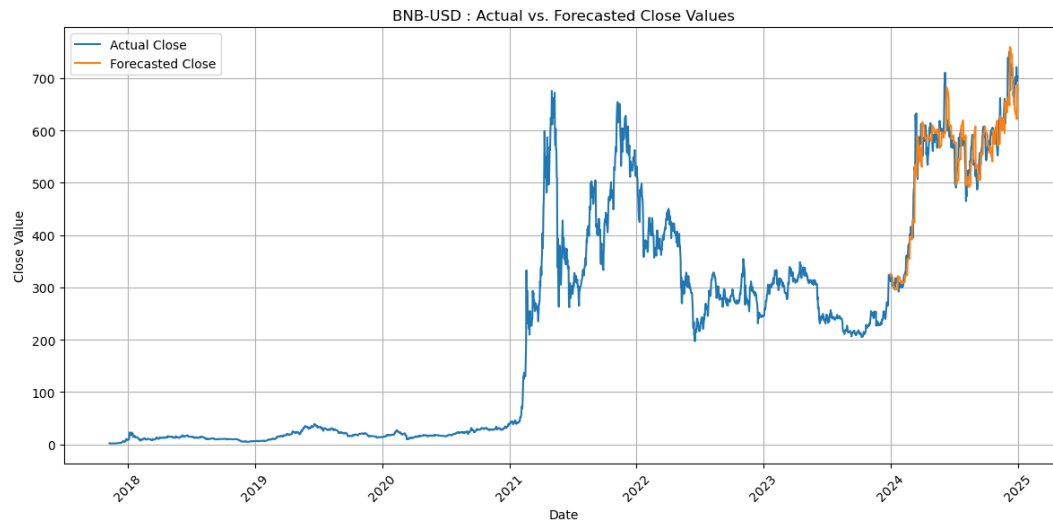


Experiment Results – TimesFM - II

Unscaled | Model: TimesFM | Horizon: 7 | Context Length: 512

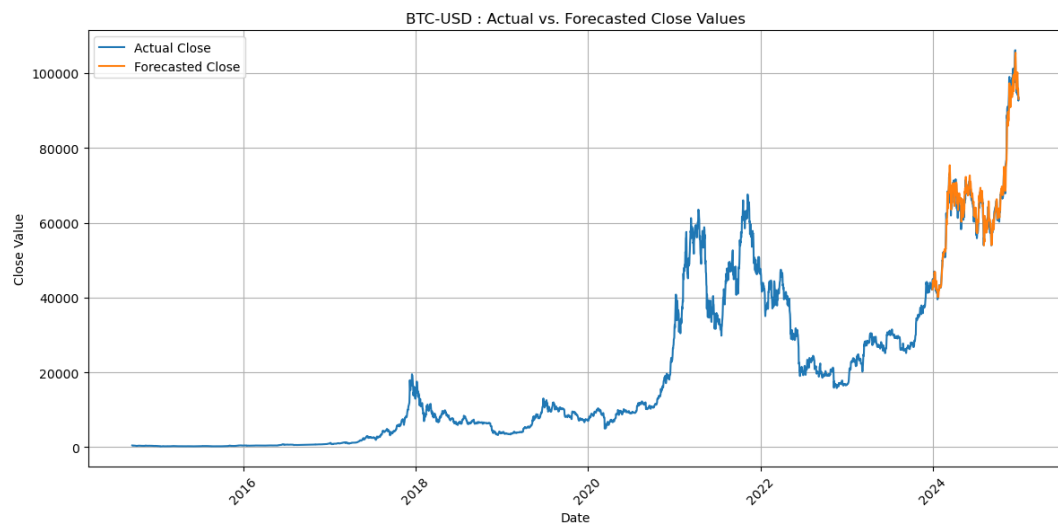


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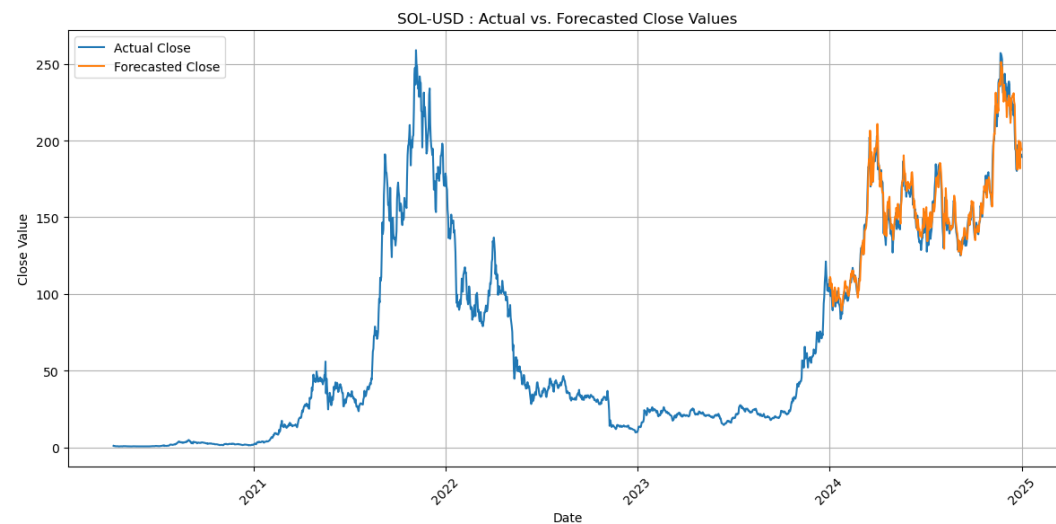
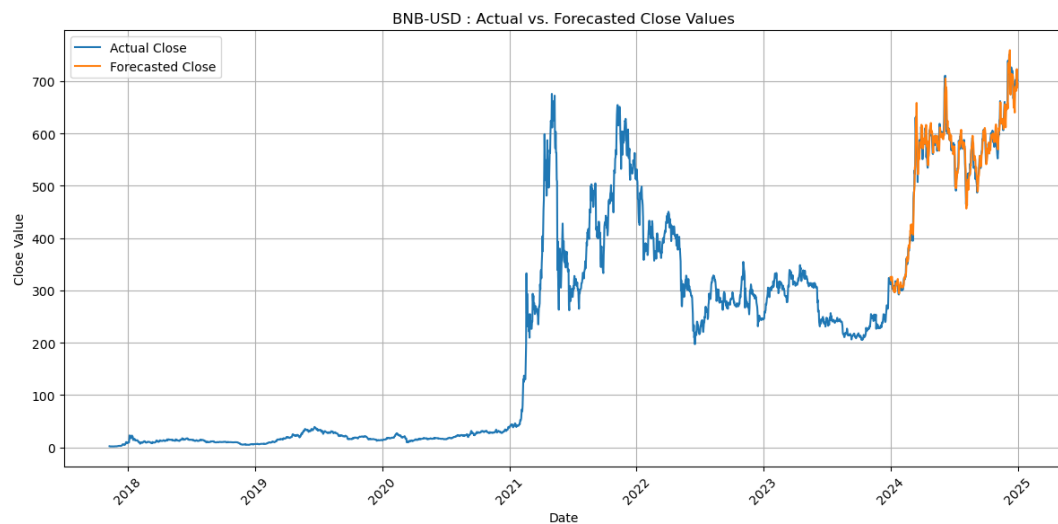


Experiment Results – TimesFM - III

Unscaled | Model: TimesFM | Horizon: 1 | Context Length: 512

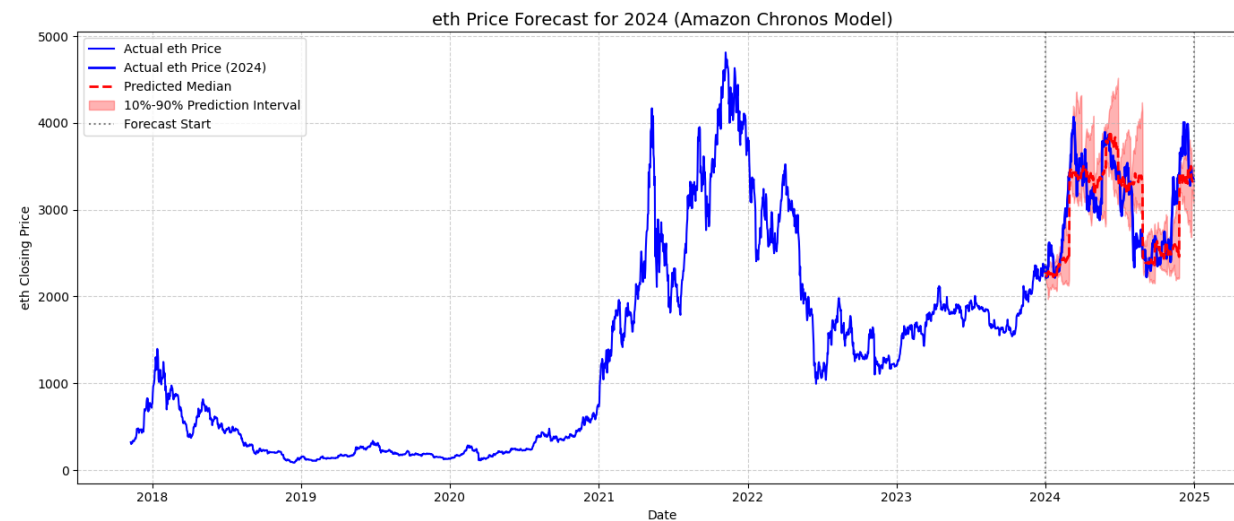
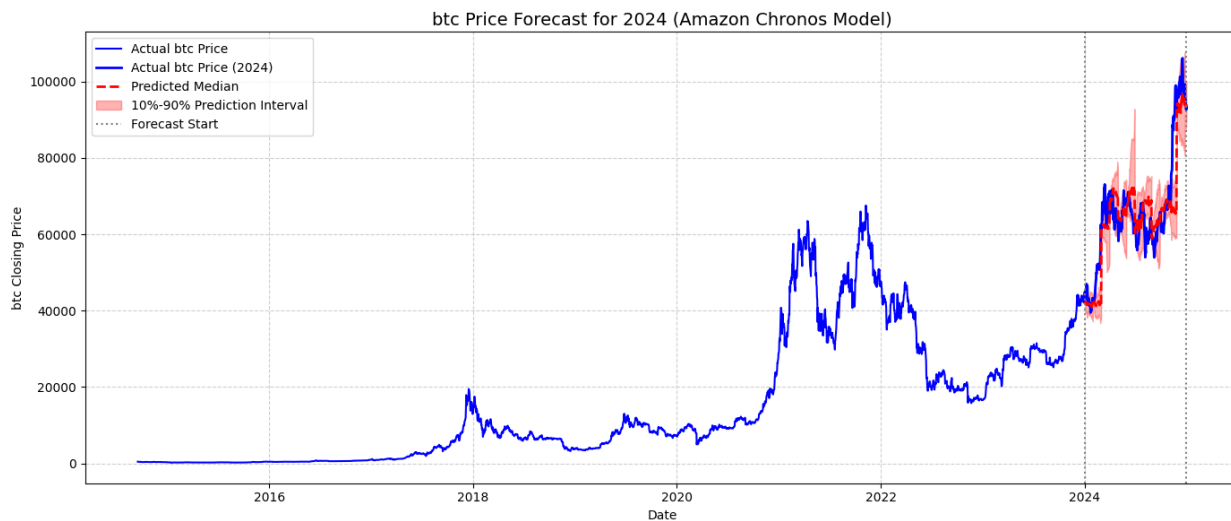


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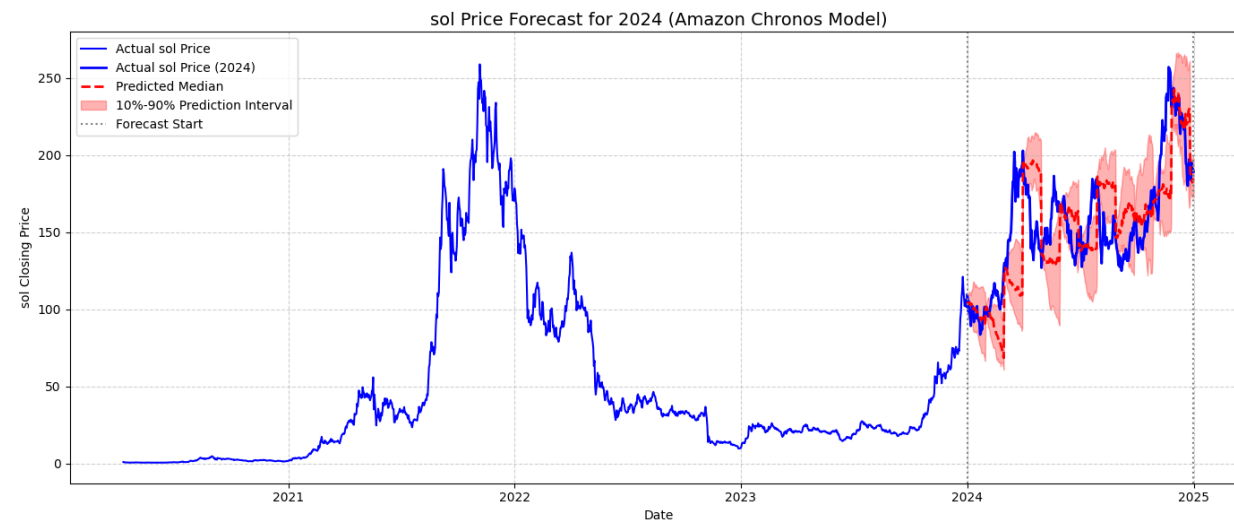
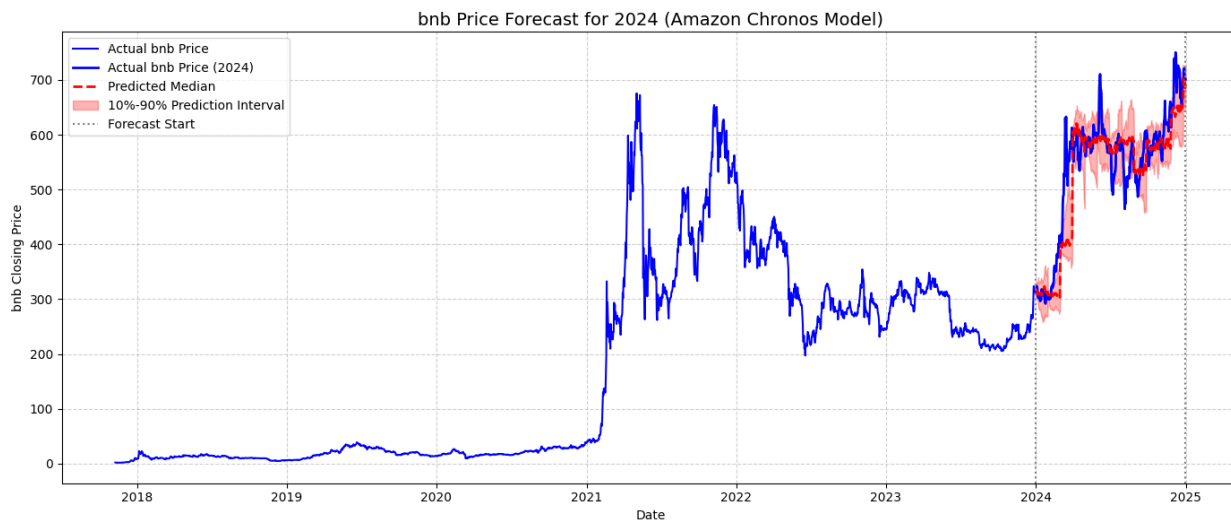


Experimental Results – Chronos - I

Unscaled | Model: Chronos | Horizon: 30 | Context Length: 512



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Experiment Results – Chronos - II

Unscaled | Model: Chronos | Horizon: 7 | Context Length: 512

btc Price Forecast for 2024 (Amazon Chronos Model)



eth Price Forecast for 2024 (Amazon Chronos Model)



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bnb Price Forecast for 2024 (Amazon Chronos Model)



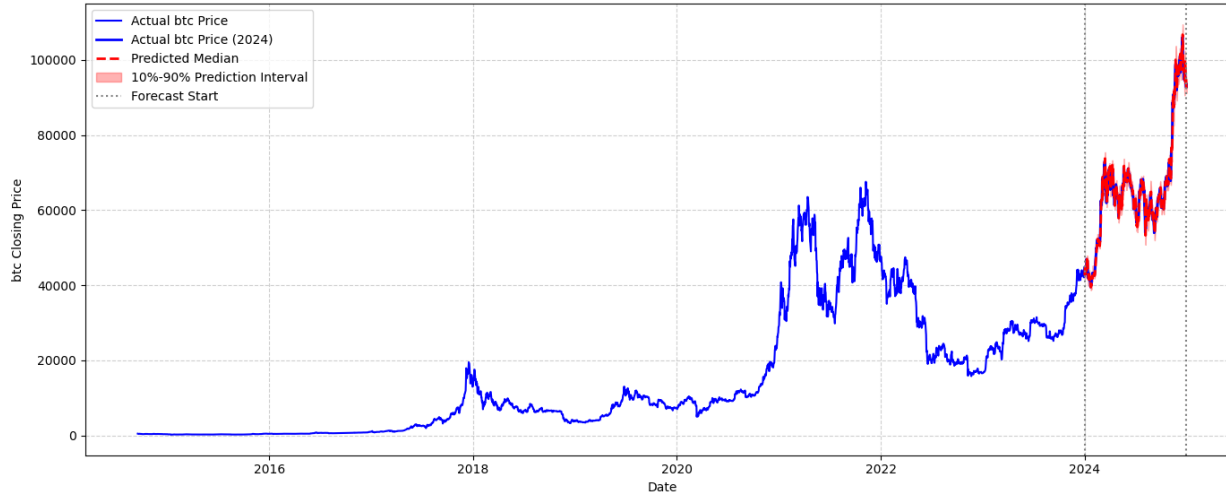
sol Price Forecast for 2024 (Amazon Chronos Model)



Experiment Results – Chronos - III

Unscaled | Model: Chronos | Horizon: 1 | Context Length: 512

btc Price Forecast for 2024 (Amazon Chronos Model)



eth Price Forecast for 2024 (Amazon Chronos Model)



Mind the timeline. Its is different for each currency

bnb Price Forecast for 2024 (Amazon Chronos Model)

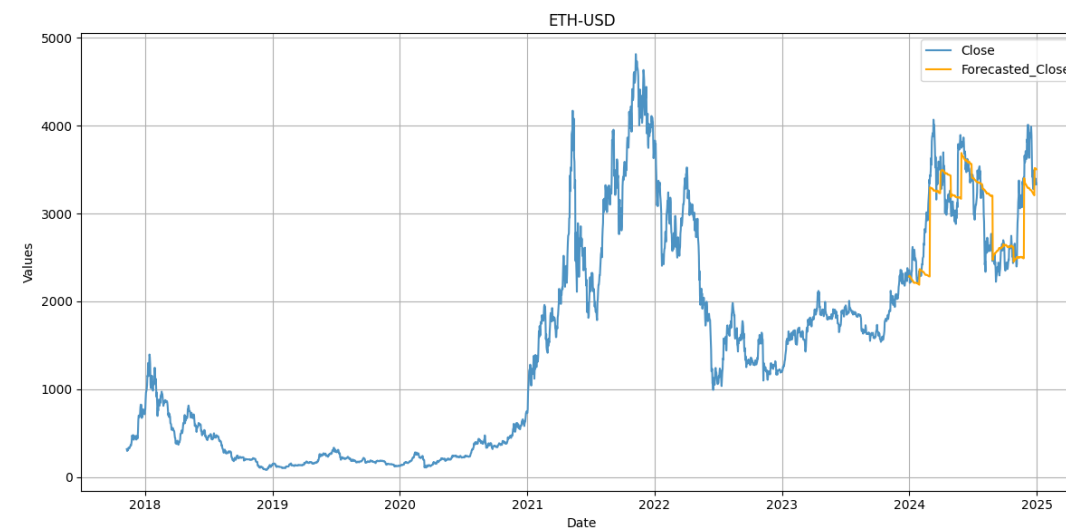
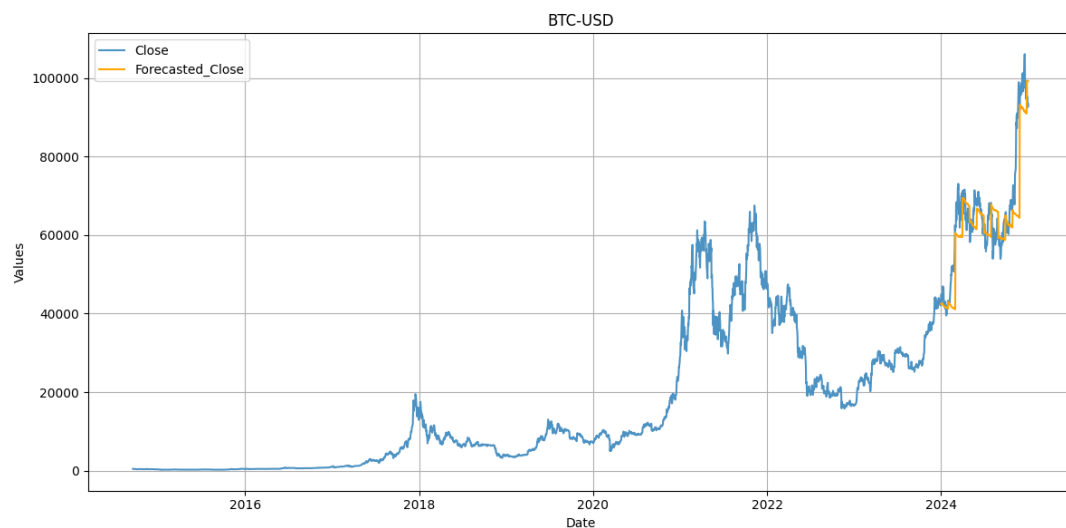


sol Price Forecast for 2024 (Amazon Chronos Model)

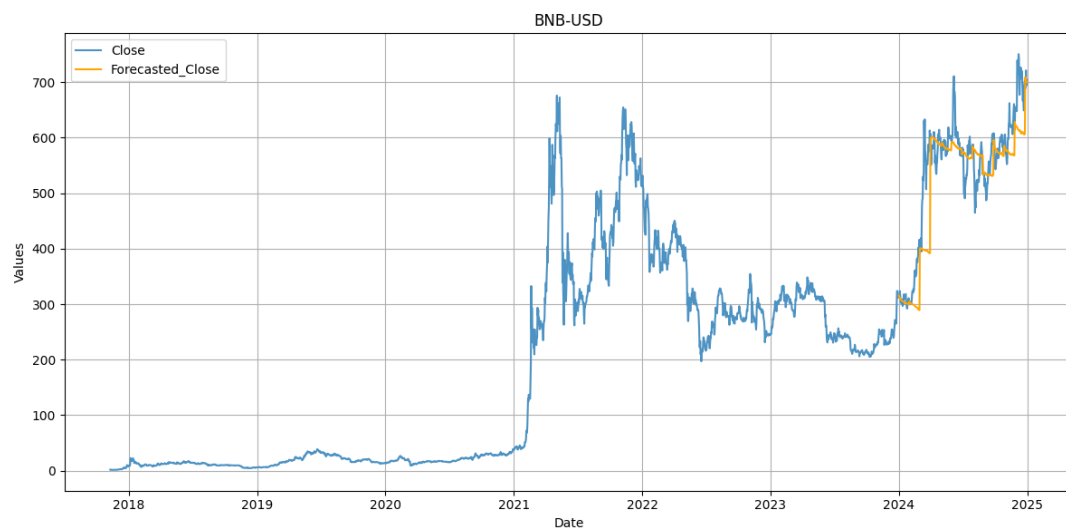


Experiment Results – TTM - I

Unscaled | Model: TTM | Horizon: 30 | Context Length: 512

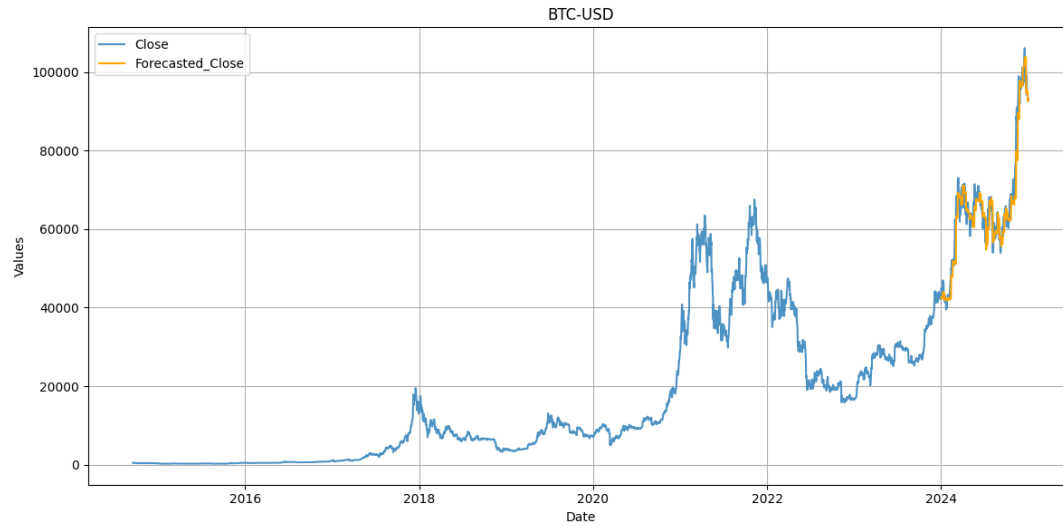


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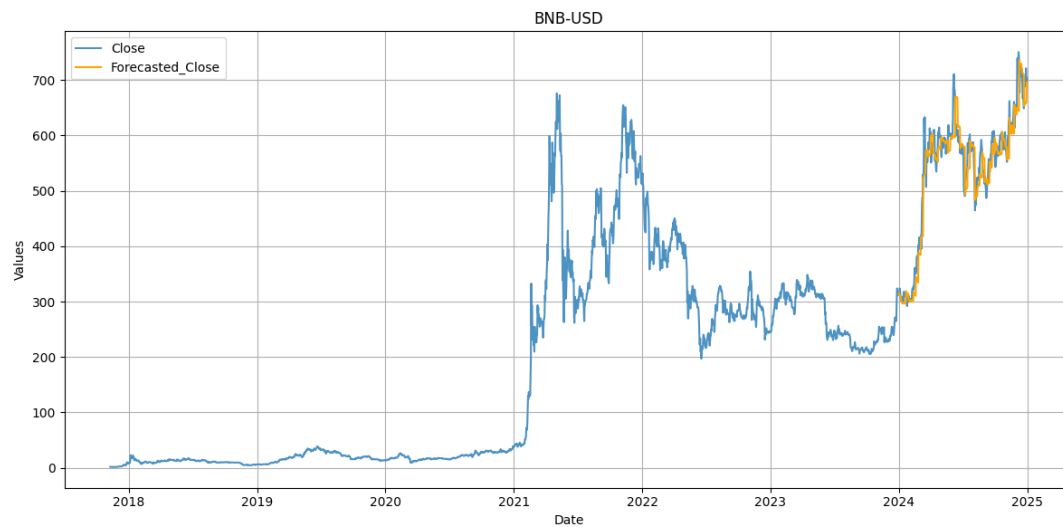


Experiment Results – TTM - II

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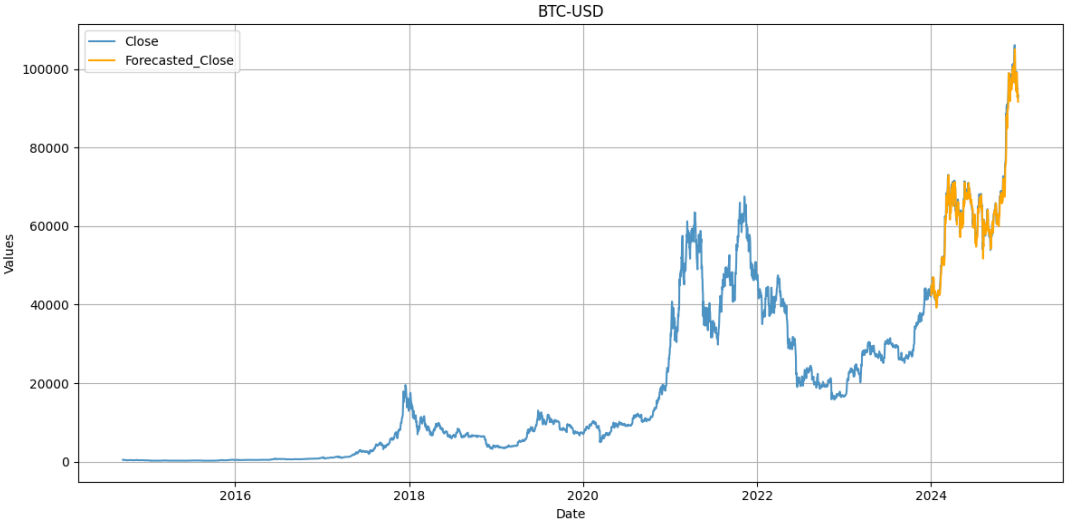


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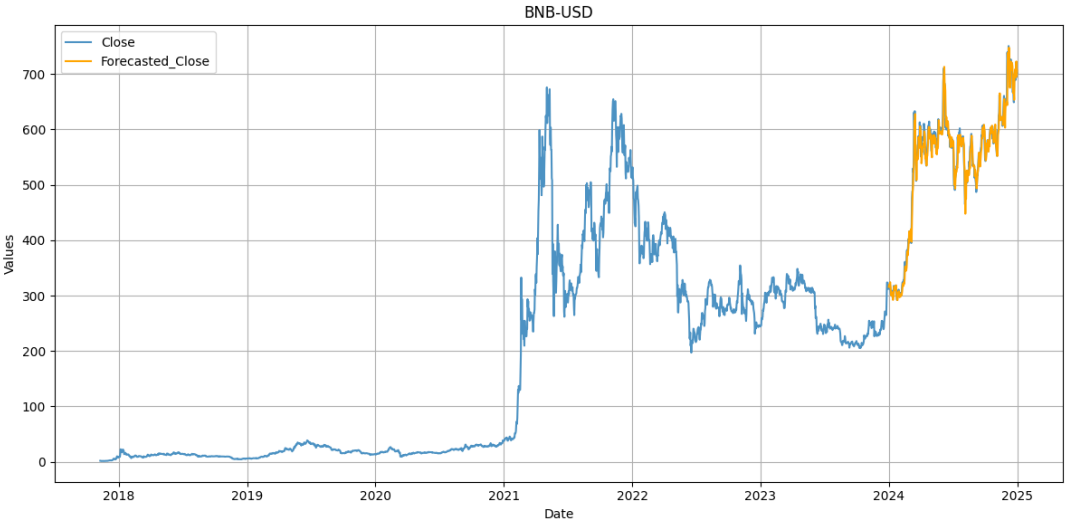


Experiment Results – TTM - III

Unscaled | Model: TTM | Horizon: 1 | Context Length: 512

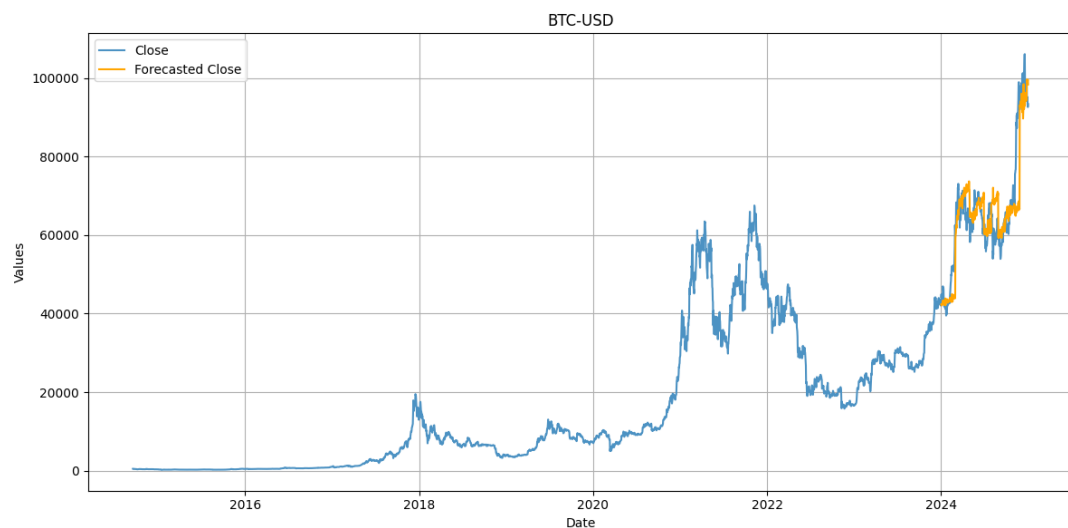


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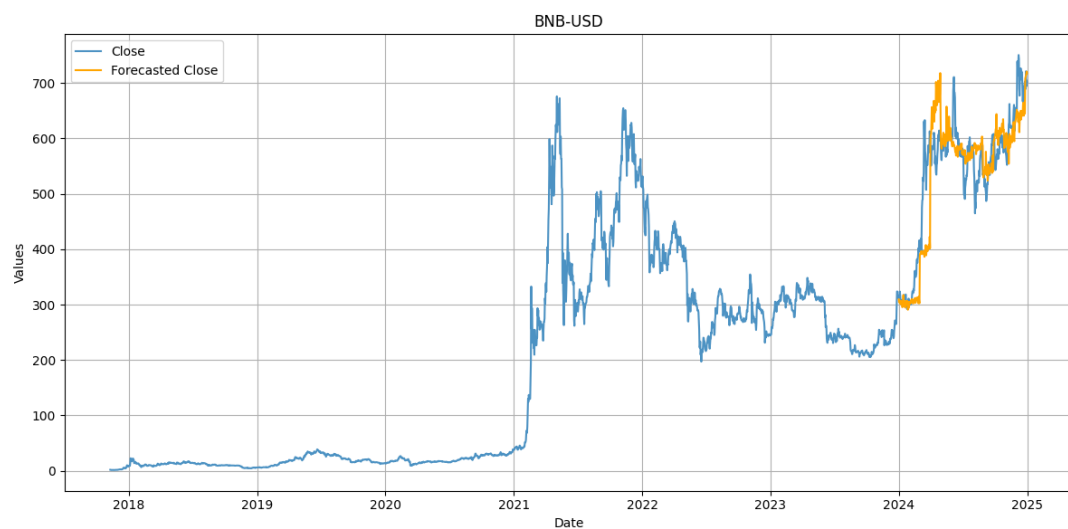


Experiment Results – Moirai - I

Unscaled | Model: Moirai | Horizon: 30 | Context Length: 512

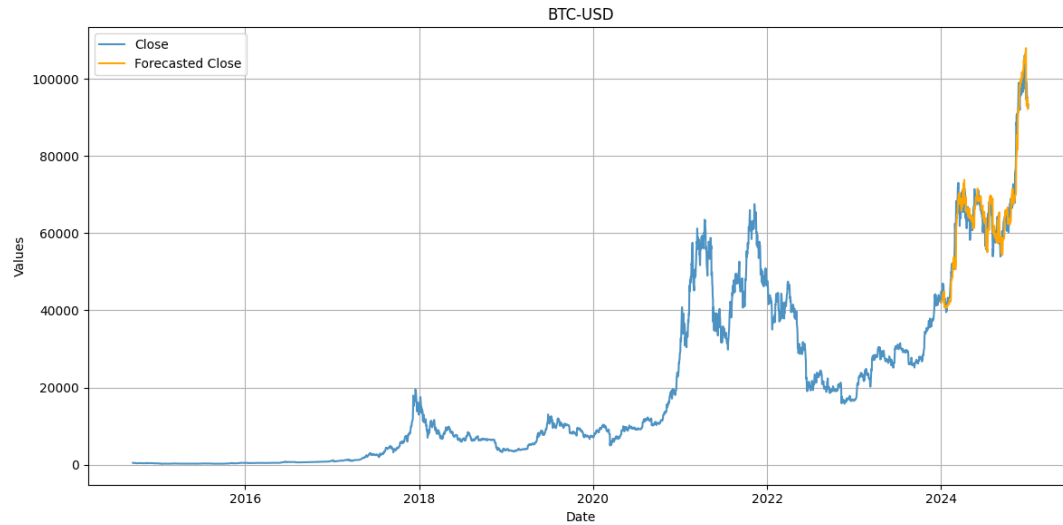


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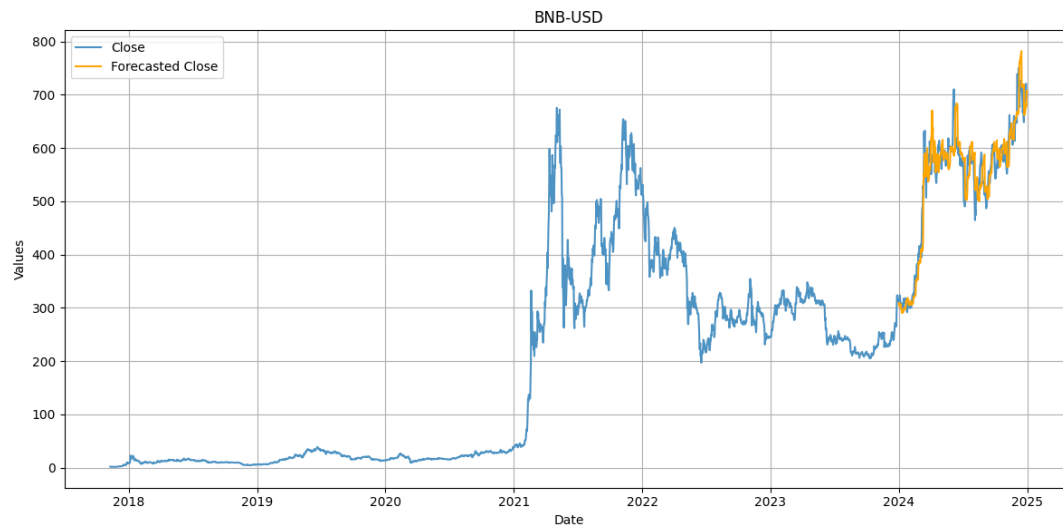


Experiment Results – Morai - II

Unscaled | Model: Moirai | Horizon: 7 | Context Length: 512

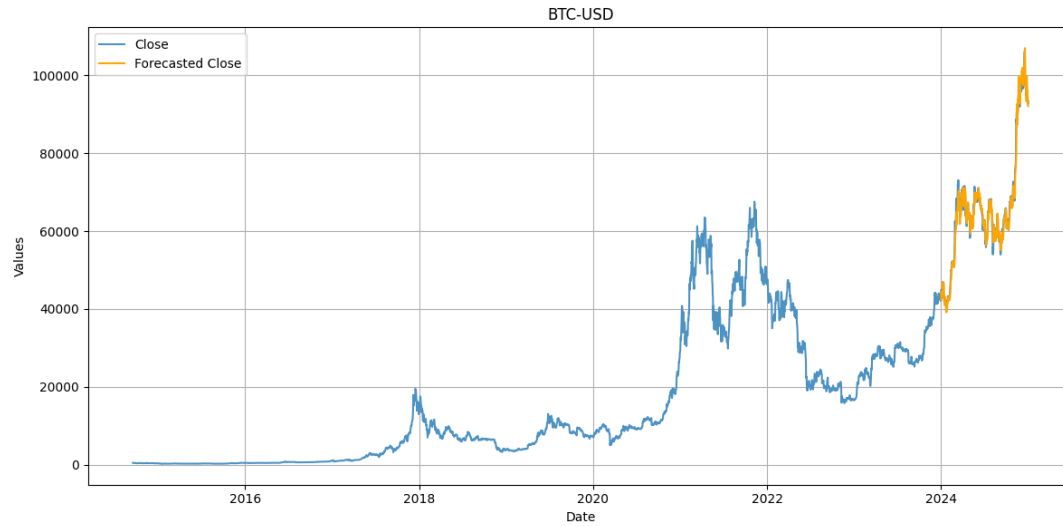


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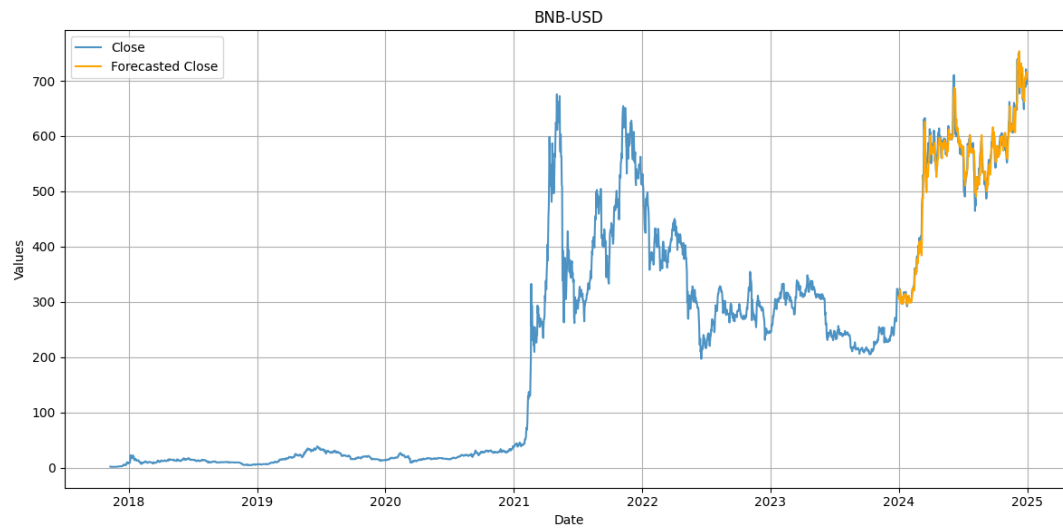


Experiment Results – Moirai - III

Unscaled | Model: Moirai | Horizon: 1 | Context Length: 512

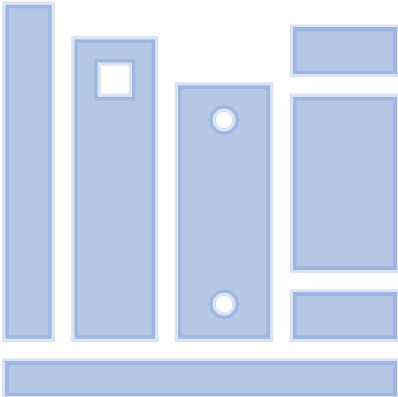


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Evaluation Metrics Table – TimesFM – I

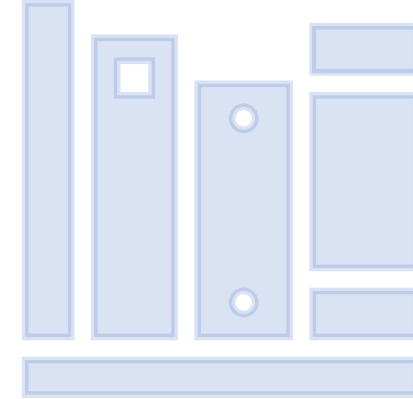
BTC						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TimesFM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	7.4226	11.7969	30.7857	8.6567	5.0090	2.5364
ME	1742.8784	3239.4415	20479.2714	1181.6526	75.7273	-305.0041
MAE	5090.2173	8221.0944	20479.2714	5763.2598	3295.2364	1688.0507
MPE	1.9162	3.5666	30.7857	0.9924	-0.0937	-0.6223
RMSE	7402.2393	11120.5600	22602.2932	7723.1694	4176.6292	2211.5484
R	0.8719	0.7148	0.7598	0.8565	0.9589	0.9889
Scalar Product	-153547.9727	58076841.3236	-22490077.3657	12411252.1913	175626978.5419	-22417878.2082
Return Score	0.5507	0.5479	0.4904	0.5096	0.4986	0.4877
Long Return	0.6474	0.6105	0.5000	0.6211	0.5421	0.5158
Short Return	0.4457	0.4800	0.4800	0.3886	0.4514	0.4571
Mean Directional Accuracy	0.5507	0.5479	0.4904	0.5096	0.4986	0.4877
Mean Directional Accuracy Positive	0.6474	0.6105	0.5000	0.6211	0.5421	0.5158
Mean Directional Accuracy Negative	0.4457	0.4800	0.4800	0.3886	0.4514	0.4571



BNB						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TimesFM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	6.8250	10.8951	29.2636	8.0115	5.1673	2.4018
ME	13.8277	28.0727	156.1461	7.5996	3.9347	-0.8367
MAE	37.3091	56.9388	156.1461	44.5397	28.6878	13.0975
MPE	2.2301	5.4607	29.2636	0.9084	0.5244	-0.2649
RMSE	56.8922	81.6522	175.3026	63.0131	37.3238	18.7054
R	0.8786	0.8119	0.7607	0.8440	0.9439	0.9858
Scalar Product	-38.5595	1924.4099	-957.1714	-3356.5184	-12839.4087	-2820.1170
Return Score	0.4466	0.5233	0.5096	0.4712	0.4795	0.4712
Long Return	0.4188	0.6126	0.4764	0.4607	0.4293	0.5236
Short Return	0.4770	0.4253	0.5460	0.4828	0.5345	0.4138
Mean Directional Accuracy	0.4466	0.5233	0.5096	0.4712	0.4795	0.4712
Mean Directional Accuracy Positive	0.4188	0.6126	0.4764	0.4607	0.4293	0.5236
Mean Directional Accuracy Negative	0.4770	0.4253	0.5460	0.4828	0.5345	0.4138

Evaluation Metrics Table – TimesFM – II

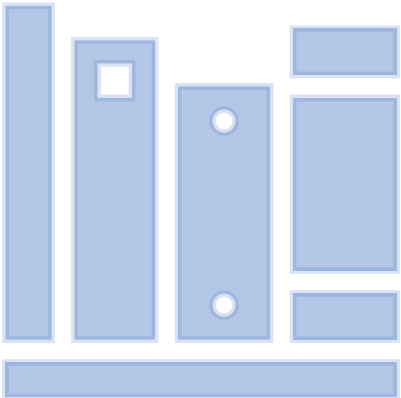
ETH						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TimesFM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	9.0057	14.5012	20.8307	12.7360	6.4317	2.9201
ME	28.9728	91.5595	487.7298	1.6122	-17.1627	-29.4584
MAE	276.1056	459.9895	657.6422	385.3562	196.3419	89.0837
MPE	0.2611	1.3732	13.9636	-1.5181	-0.8669	-0.9998
RMSE	364.3019	584.0289	787.5722	517.3712	255.2013	118.7584
R	0.7427	0.3544	0.1328	0.4553	0.8771	0.9751
Scalar Product	24408.6257	27924.4320	-9101.4936	-26698.6671	354544.7348	-49548.8107
Return Score	0.2329	0.4877	0.4986	0.4630	0.4822	0.4548
Long Return	0.2304	0.4921	0.4869	0.4555	0.5026	0.4921
Short Return	0.2356	0.4828	0.5115	0.4713	0.4598	0.4138
Mean Directional Accuracy	0.2329	0.4877	0.4986	0.4630	0.4822	0.4548
Mean Directional Accuracy Positive	0.2304	0.4921	0.4869	0.4555	0.5026	0.4921
Mean Directional Accuracy Negative	0.2356	0.4828	0.5115	0.4713	0.4598	0.4138



SOL						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TimesFM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	13.1475	24.3638	35.0459	14.9781	6.7310	3.9940
ME	-0.5452	7.2170	20.6437	-0.4802	-0.7168	-1.5554
MAE	21.1164	38.4662	50.0925	24.0087	10.4488	6.0991
MPE	-1.8483	3.2603	15.4646	-1.8105	-0.7974	-1.3432
RMSE	28.7276	49.2536	58.5803	32.6841	15.1125	7.8043
R	0.7011	0.4504	0.6067	0.6547	0.9203	0.9787
Scalar Product	1478.4553	1083.0943	-114.6795	1910.9223	379.1411	-981.4638
Return Score	0.3342	0.5178	0.5014	0.4603	0.5315	0.5205
Long Return	0.3370	0.5856	0.5635	0.4088	0.5359	0.5193
Short Return	0.3315	0.4511	0.4402	0.5109	0.5272	0.5217
Mean Directional Accuracy	0.3342	0.5178	0.5014	0.4603	0.5315	0.5205
Mean Directional Accuracy Positive	0.3370	0.5856	0.5635	0.4088	0.5359	0.5193
Mean Directional Accuracy Negative	0.3315	0.4511	0.4402	0.5109	0.5272	0.5217

Evaluation Metrics Table – Chronos – I

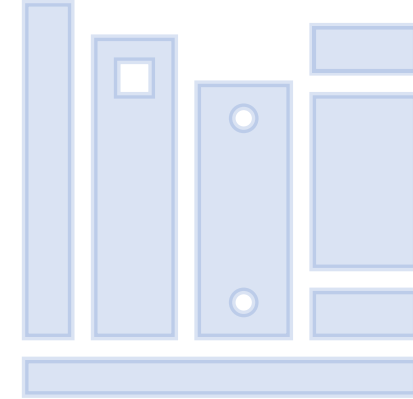
BTC						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Chronos		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	7.4226	11.7969	30.7857	7.4117	4.5537	2.0218
ME	1742.8784	3239.4415	20479.2714	775.8249	128.4270	41.3087
MAE	5090.2173	8221.0944	20479.2714	4963.0446	2982.3971	1335.1071
MPE	1.9162	3.5666	30.7857	0.5220	0.1772	0.0508
RMSE	7402.2393	11120.5600	22602.2932	7184.7332	3848.7169	1859.8386
R	0.8719	0.7148	0.7598	0.8758	0.9669	0.9920
Scalar Product	-153547.9727	58076841.3236	-22490077.3657	31949163.8925	168698375.7376	-22149873.0630
Return Score	0.5507	0.5479	0.4904	0.4822	0.4329	0.4932
Long Return	0.6474	0.6105	0.5000	0.5474	0.5105	0.5211
Short Return	0.4457	0.4800	0.4800	0.4114	0.3486	0.4629
Mean Directional Accuracy	0.5507	0.5479	0.4904	0.4822	0.4329	0.4932
Mean Directional Accuracy Positive	0.6474	0.6105	0.5000	0.5474	0.5105	0.5211
Mean Directional Accuracy Negative	0.4457	0.4800	0.4800	0.4114	0.3486	0.4629



BNB						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Chronos		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	6.8250	10.8951	29.2636	6.9344	4.7503	2.2077
ME	13.8277	28.0727	156.1461	14.4979	3.9087	0.7531
MAE	37.3091	56.9388	156.1461	37.6983	26.2250	12.0669
MPE	2.2301	5.4607	29.2636	2.3339	0.7112	0.1444
RMSE	56.8922	81.6522	175.3026	56.0816	34.2638	17.2535
R	0.8786	0.8119	0.7607	0.8820	0.9545	0.9881
Scalar Product	-38.5595	1924.4099	-957.1714	1967.1346	-10417.7316	-3003.0220
Return Score	0.4466	0.5233	0.5096	0.4740	0.4356	0.4630
Long Return	0.4188	0.6126	0.4764	0.4712	0.4660	0.4974
Short Return	0.4770	0.4253	0.5460	0.4770	0.4023	0.4253
Mean Directional Accuracy	0.4466	0.5233	0.5096	0.4740	0.4356	0.4630
Mean Directional Accuracy Positive	0.4188	0.6126	0.4764	0.4712	0.4660	0.4974
Mean Directional Accuracy Negative	0.4770	0.4253	0.5460	0.4770	0.4023	0.4253

Evaluation Metrics Table – Chronos – II

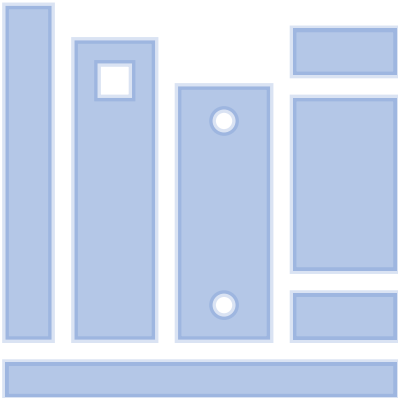
ETH						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Chronos		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	9.0057	14.5012	20.8307	10.0172	5.6979	2.5137
ME	28.9728	91.5595	487.7298	-9.9776	5.5854	0.1528
MAE	276.1056	459.9895	657.6422	308.5588	173.0978	76.2969
MPE	0.2611	1.3732	13.9636	-0.8324	0.0577	-0.0227
RMSE	364.3019	584.0289	787.5722	390.7390	224.9953	105.5331
R	0.7427	0.3544	0.1328	0.7349	0.9080	0.9788
Scalar Product	24408.6257	27924.4320	-9101.4936	6950.8020	349407.6724	-37458.7236
Return Score	0.2329	0.4877	0.4986	0.4822	0.4219	0.4986
Long Return	0.2304	0.4921	0.4869	0.5183	0.4555	0.5288
Short Return	0.2356	0.4828	0.5115	0.4425	0.3851	0.4655
Mean Directional Accuracy	0.2329	0.4877	0.4986	0.4822	0.4219	0.4986
Mean Directional Accuracy Positive	0.2304	0.4921	0.4869	0.5183	0.4555	0.5288
Mean Directional Accuracy Negative	0.2356	0.4828	0.5115	0.4425	0.3851	0.4655



SOL						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Chronos		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	13.1475	24.3638	35.0459	15.7908	7.0260	3.4104
ME	-0.5452	7.2170	20.6437	-0.1087	0.6287	0.0144
MAE	21.1164	38.4662	50.0925	25.1916	10.9984	5.2570
MPE	-1.8483	3.2603	15.4646	-1.3189	0.4503	-0.0180
RMSE	28.7276	49.2536	58.5803	33.2043	14.5872	6.9559
R	0.7011	0.4504	0.6067	0.6583	0.9312	0.9828
Scalar Product	1478.4553	1083.0943	-114.6795	1656.2266	731.5922	-568.2265
Return Score	0.3342	0.5178	0.5014	0.4493	0.5342	0.5425
Long Return	0.3370	0.5856	0.5635	0.3646	0.4751	0.5304
Short Return	0.3315	0.4511	0.4402	0.5326	0.5924	0.5543
Mean Directional Accuracy	0.3342	0.5178	0.5014	0.4493	0.5342	0.5425
Mean Directional Accuracy Positive	0.3370	0.5856	0.5635	0.3646	0.4751	0.5304
Mean Directional Accuracy Negative	0.3315	0.4511	0.4402	0.5326	0.5924	0.5543

Evaluation Metrics Table – Tiny Time Mixers – I

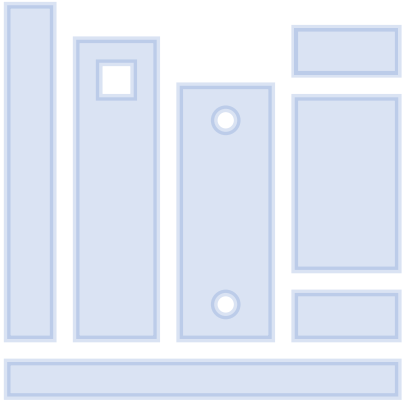
BTC						
Type	BenchmarkingTraditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TTM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	7.4226	11.7969	30.7857	7.8205	4.6689	2.1309
ME	1742.8784	3239.4415	20479.2714	3152.0168	8.8385	470.8715
MAE	5090.2173	8221.0944	20479.2714	5447.2776	25.7264	1406.7461
MPE	1.9162	3.5666	30.7857	4.0519	1.5735	0.6405
RMSE	7402.2393	11120.5600	22602.2932	8017.2147	34.3992	1957.2723
R	0.8719	0.7148	0.7598	0.8648	0.9554	0.9915
Scalar Product	-153547.9727	58076841.3236	-22490077.3657	-7264848.1110	-10257.1327	-52602085.2800
Return Score	0.5507	0.5479	0.4904	0.4821	0.5041	0.4684
Long Return	0.6474	0.6105	0.5000	0.2684	0.3979	0.5000
Short Return	0.4457	0.4800	0.4800	0.7142	0.6206	0.4342
Mean Directional Accuracy	0.5507	0.5479	0.4904	0.4821	0.5041	0.4684
Mean Directional Accuracy Positive	0.6474	0.6105	0.5000	0.2684	0.3979	0.5000
Mean Directional Accuracy Negative	0.4457	0.4800	0.4800	0.7142	0.6206	0.4342



BNB						
Type	BenchmarkingTraditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TTM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	6.8250	10.8951	29.2636	7.3106	4.6689	2.2552
ME	13.8277	28.0727	156.1461	24.0157	8.8386	3.5878
MAE	37.3091	56.9388	156.1461	40.3955	25.7265	12.3987
MPE	2.2301	5.4607	29.2636	4.1458	1.5735	0.6385
RMSE	56.8922	81.6522	175.3026	61.1665	34.3992	17.5129
R	0.8786	0.8119	0.7607	0.8722	0.9554	0.9881
Scalar Product	-38.5595	1924.4099	-957.1714	-1107.8017	-10257.1327	-319.9691
Return Score	0.4466	0.5233	0.5096	0.4767	0.5041	0.4795
Long Return	0.4188	0.6126	0.4764	0.2723	0.3979	0.5079
Short Return	0.4770	0.4253	0.5460	0.7011	0.6207	0.4483
Mean Directional Accuracy	0.4466	0.5233	0.5096	0.4767	0.5041	0.4795
Mean Directional Accuracy Positive	0.4188	0.6126	0.4764	0.2723	0.3979	0.5079
Mean Directional Accuracy Negative	0.4770	0.4253	0.5460	0.7011	0.6207	0.4483

Evaluation Metrics Table – Tiny Time Mixers – II

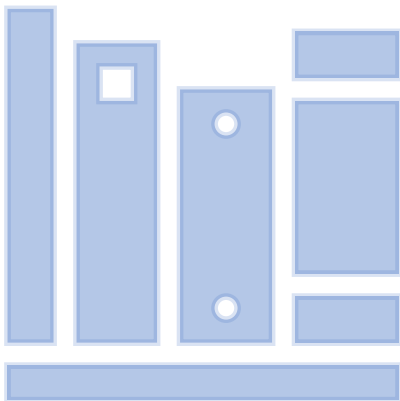
ETH						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TTM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	9.0057	14.5012	20.8307	9.2948	5.7203	2.6897
ME	28.9728	91.5595	487.7298	64.6087	19.9946	16.5108
MAE	276.1056	459.9895	657.6422	285.6775	174.6830	82.0190
MPE	0.2611	1.3732	13.9636	1.2439	0.3495	0.4286
RMSE	364.3019	584.0289	787.5722	373.8621	233.5284	111.1645
R	0.7427	0.3544	0.1328	0.7179	0.8941	0.9764
Scalar Product	24408.6257	27924.4320	-9101.4936	-46525.0048	241688.1080	9100.2000
Return Score	0.2329	0.4877	0.4986	0.4685	0.5205	0.4767
Long Return	0.2304	0.4921	0.4869	0.3141	0.4607	0.4974
Short Return	0.2356	0.4828	0.5115	0.6379	0.5862	0.4540
Mean Directional Accuracy	0.2329	0.4877	0.4986	0.4685	0.5205	0.4767
Mean Directional Accuracy Positive	0.2304	0.4921	0.4869	0.3141	0.4607	0.4974
Mean Directional Accuracy Negative	0.2356	0.4828	0.5115	0.6379	0.5862	0.4540



SOL						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	TTM		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	13.1475	24.3638	35.0459	12.9182	6.7618	3.3896
ME	-0.5452	7.2170	20.6437	1.5768	0.8821	0.9010
MAE	21.1164	38.4662	50.0925	20.9523	10.6239	5.2748
MPE	-1.8483	3.2603	15.4646	-0.5603	0.3990	0.4641
RMSE	28.7276	49.2536	58.5803	29.1340	14.4369	7.1348
R	0.7011	0.4504	0.6067	0.6877	0.9281	0.9818
Scalar Product	1478.4553	1083.0943	-114.6795	1525.8444	476.3130	-883.3095
Return Score	0.3342	0.5178	0.5014	0.4986	0.5616	0.5233
Long Return	0.3370	0.5856	0.5635	0.4309	0.5635	0.5304
Short Return	0.3315	0.4511	0.4402	0.5652	0.5598	0.5163
Mean Directional Accuracy	0.3342	0.5178	0.5014	0.4986	0.5616	0.5233
Mean Directional Accuracy Positive	0.3370	0.5856	0.5635	0.4309	0.5635	0.5304
Mean Directional Accuracy Negative	0.3315	0.4511	0.4402	0.5652	0.5598	0.5163

Evaluation Metrics Table – Moirai – I

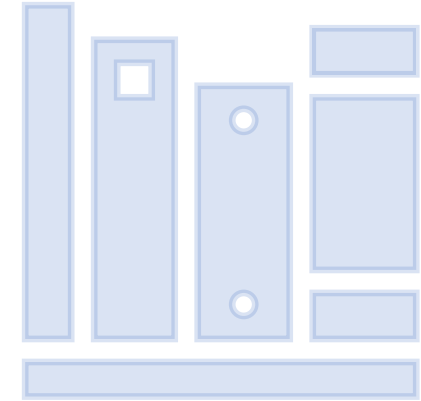
BTC						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Moirai		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	7.4226	11.7969	30.7857	7.8042	4.5633	2.3044
ME	1742.8784	3239.4415	20479.2714	7.7342	238.6896	239.1631
MAE	5090.2173	8221.0944	20479.2714	42.8013	2996.6965	1518.4464
MPE	1.9162	3.5666	30.7857	1.3692	0.3868	0.3780
RMSE	7402.2393	11120.5600	22602.2932	61.8201	3834.2699	2095.1794
R	0.8719	0.7148	0.7598	0.8639	0.9677	0.9901
Scalar Product	-153547.9727	58076841.3236	-22490077.3657	8270.6943	120321755.1463	#####
Return Score	0.5507	0.5479	0.4904	0.5232	0.4794	0.4712
Long Return	0.6474	0.6105	0.5000	0.5497	0.5894	0.4894
Short Return	0.4457	0.4800	0.4800	0.4942	0.3600	0.4514
Mean Directional Accuracy	0.5507	0.5479	0.4904	0.5232	0.4794	0.4712
Mean Directional Accuracy Positive	0.6474	0.6105	0.5000	0.5497	0.5894	0.4894
Mean Directional Accuracy Negative	0.4457	0.4800	0.4800	0.4942	0.3600	0.4514



BNB						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Moirai		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	6.8250	10.8951	29.2636	7.8043	4.8288	2.4266
ME	13.8277	28.0727	156.1461	7.7342	0.5272	2.4329
MAE	37.3091	56.9388	156.1461	42.8013	26.6470	13.3683
MPE	2.2301	5.4607	29.2636	1.3693	0.1288	0.4350
RMSE	56.8922	81.6522	175.3026	61.8202	35.2326	19.0754
R	0.8786	0.8119	0.7607	0.8639	0.9524	0.9855
Scalar Product	-38.5595	1924.4099	-957.1714	8270.6943	-10955.9850	-3932.5891
Return Score	0.4466	0.5233	0.5096	0.5233	0.5013	0.4739
Long Return	0.4188	0.6126	0.4764	0.5497	0.6020	0.5287
Short Return	0.4770	0.4253	0.5460	0.4943	0.3908	0.4137
Mean Directional Accuracy	0.4466	0.5233	0.5096	0.5233	0.5013	0.4739
Mean Directional Accuracy Positive	0.4188	0.6126	0.4764	0.5497	0.6020	0.5287
Mean Directional Accuracy Negative	0.4770	0.4253	0.5460	0.4943	0.3908	0.4137

Evaluation Metrics Table – Moirai – II

ETH						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Moirai		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	9.0057	14.5012	20.8307	8.9593	5.6951	2.9054
ME	28.9728	91.5595	487.7298	9.4103	11.9812	22.0545
MAE	276.1056	459.9895	657.6422	271.5315	173.7158	89.5216
MPE	0.2611	1.3732	13.9636	-0.5303	0.1425	0.5517
RMSE	364.3019	584.0289	787.5722	354.0573	228.9240	122.8087
R	0.7427	0.3544	0.1328	0.7422	0.8997	0.9715
Scalar Product	24408.6257	27924.4320	-9101.4936	110510.6670	-22556.0860	-40675.1358
Return Score	0.2329	0.4877	0.4986	0.4712	0.4849	0.4657
Long Return	0.2304	0.4921	0.4869	0.5392	0.5759	0.4554
Short Return	0.2356	0.4828	0.5115	0.3965	0.3850	0.4770
Mean Directional Accuracy	0.2329	0.4877	0.4986	0.4712	0.4849	0.4675
Mean Directional Accuracy Positive	0.2304	0.4921	0.4869	0.5392	0.5759	0.4554
Mean Directional Accuracy Negative	0.2356	0.4828	0.5115	0.3965	0.3850	0.4770

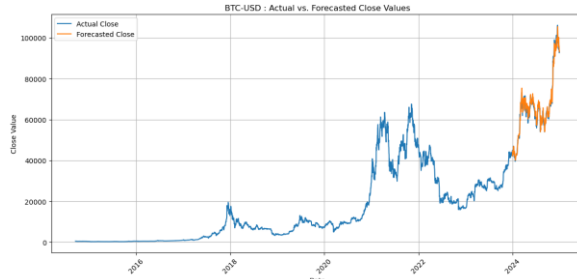


SOL						
Type	Benchmarking Traditional Models			Foundational Models		
Model	ARIMA	LSTM	Prophet	Moirai		
Horizon	30	30	30	30	7	1
Metrics						
MAPE	13.1475	24.3638	35.0459	14.6279	7.0991	14.6281
ME	-0.5452	7.2170	20.6437	-4.4630	-0.1464	10.9736
MAE	21.1164	38.4662	50.0925	22.9102	10.9736	22.9104
MPE	-1.8483	3.2603	15.4646	-4.5771	-0.1052	-4.5775
RMSE	28.7276	49.2536	58.5803	30.4252	14.8188	30.4254
R	0.7011	0.4504	0.6067	0.6773	0.9289	0.6773
Scalar Product	1478.4553	1083.0943	-114.6795	2766.2710	-203.5260	2766.2031
Return Score	0.3342	0.5178	0.5014	0.4986	0.4575	0.4986
Long Return	0.3370	0.5856	0.5635	0.5359	0.5635	0.5359
Short Return	0.3315	0.4511	0.4402	0.4619	0.3532	0.4619
Mean Directional Accuracy	0.3342	0.5178	0.5014	0.4986	0.4575	0.4986
Mean Directional Accuracy Positive	0.3370	0.5856	0.5635	0.5359	0.5635	0.5359
Mean Directional Accuracy Negative	0.3315	0.4511	0.4402	0.4619	0.3532	0.4619

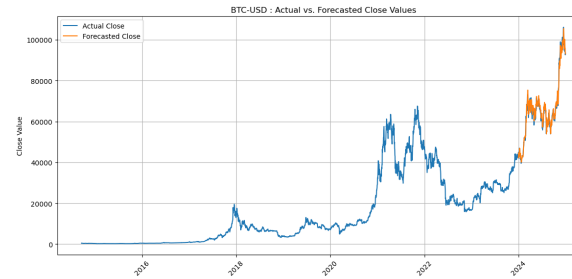
A Note on scaled vs unscaled inputs - I

Model: **TimesFM** | Horizon: 1 | Context Length: 512

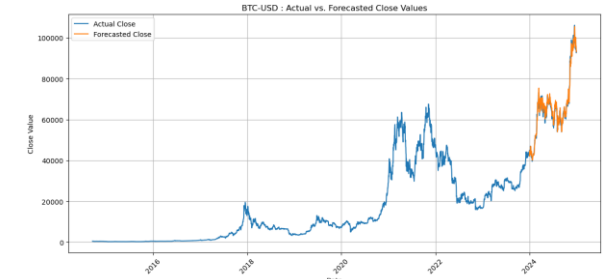
Normalized



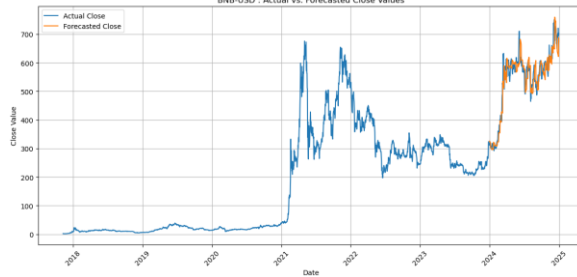
Unscaled



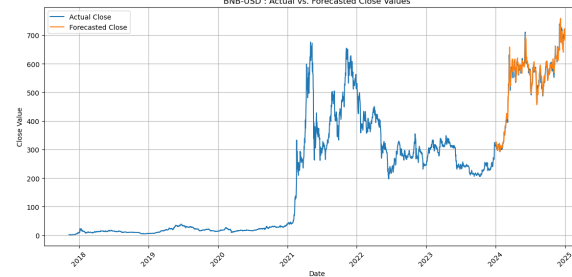
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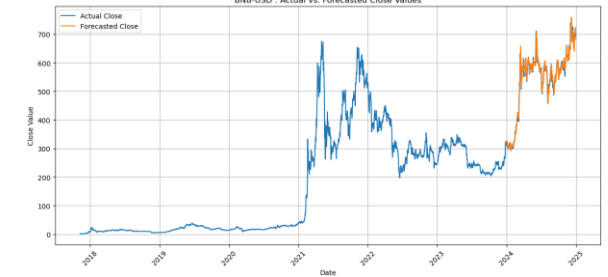
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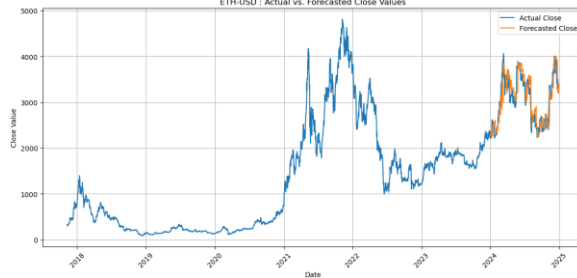
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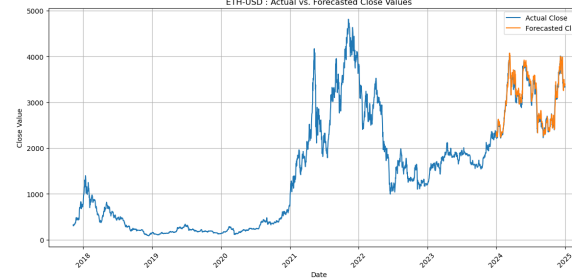
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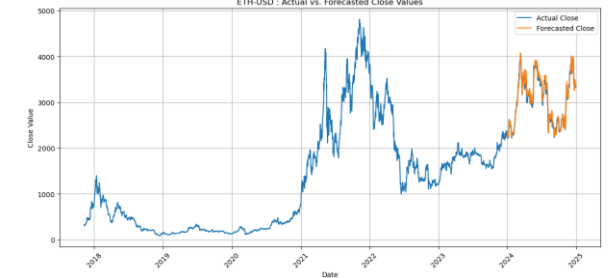
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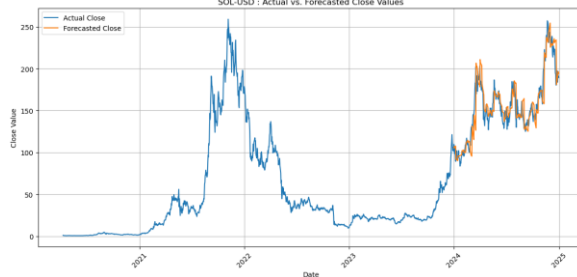
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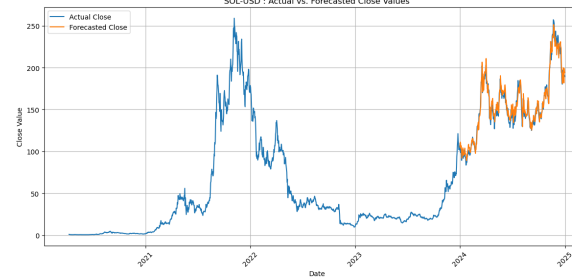
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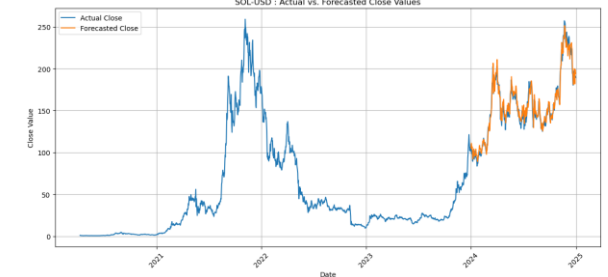
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Unscaled



Standardized



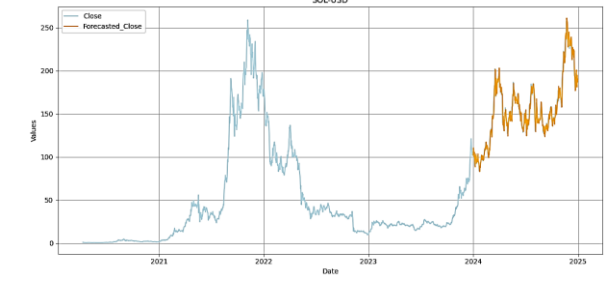
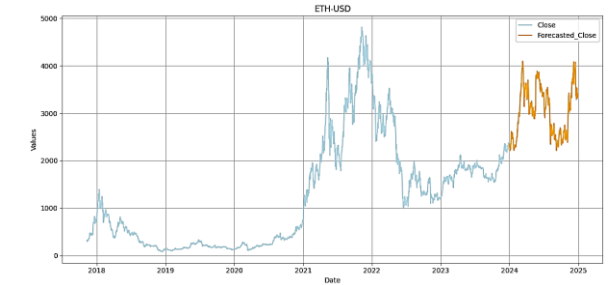
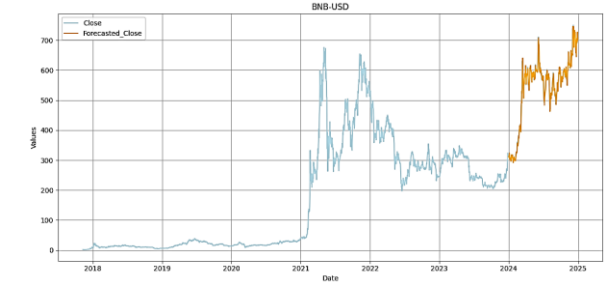
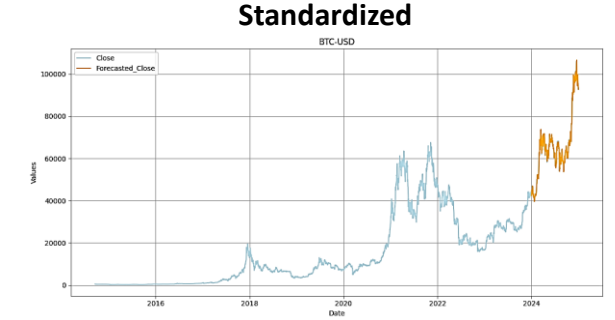
Not a major difference between scaled and unscaled inputs. Standardization seems to be better than Normalization. Scaling is Crucial for finetuning

BTC

BNB

ETH

SOL

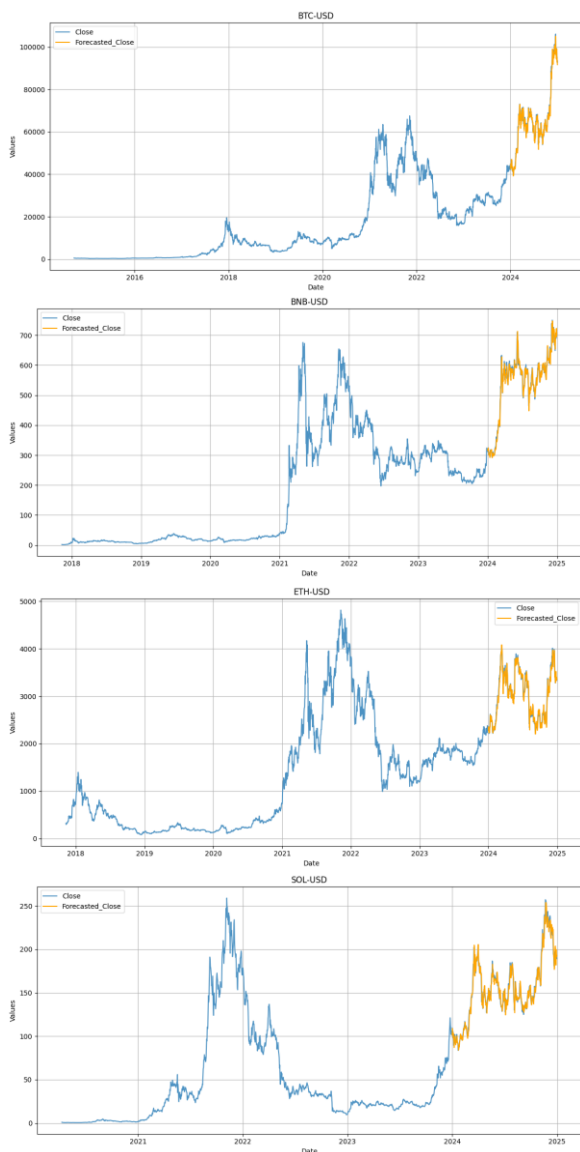


Not a major difference between scaled and unscaled inputs. Standardization seems to be better than Normalization. Scaling is Crucial for finetuning

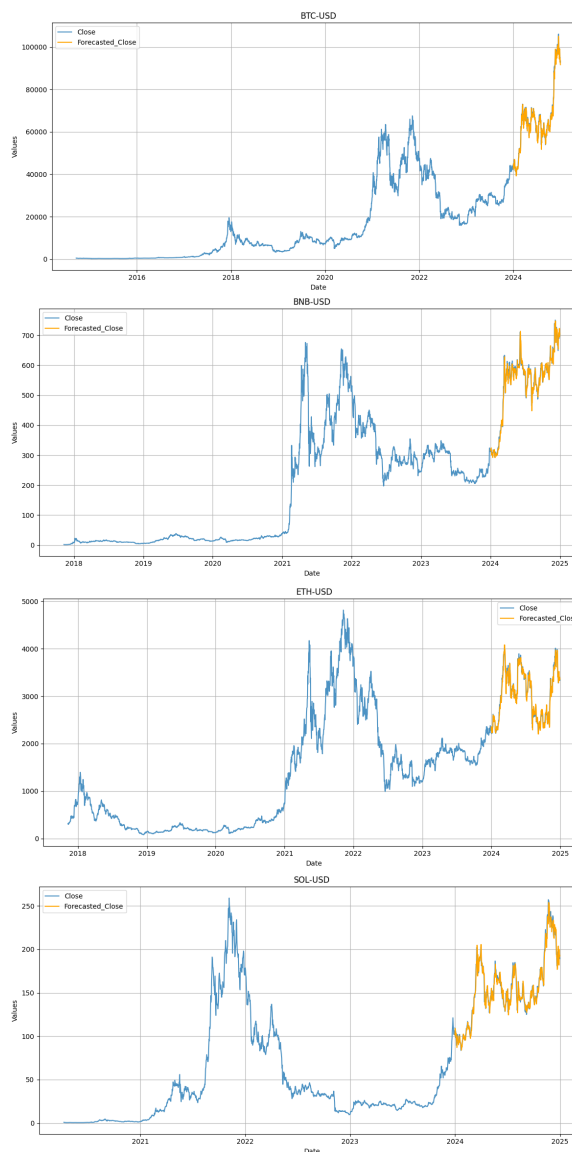
A Note on scaled vs unscaled inputs - III

Model: TTM | Horizon: 1 | Context Length: 512

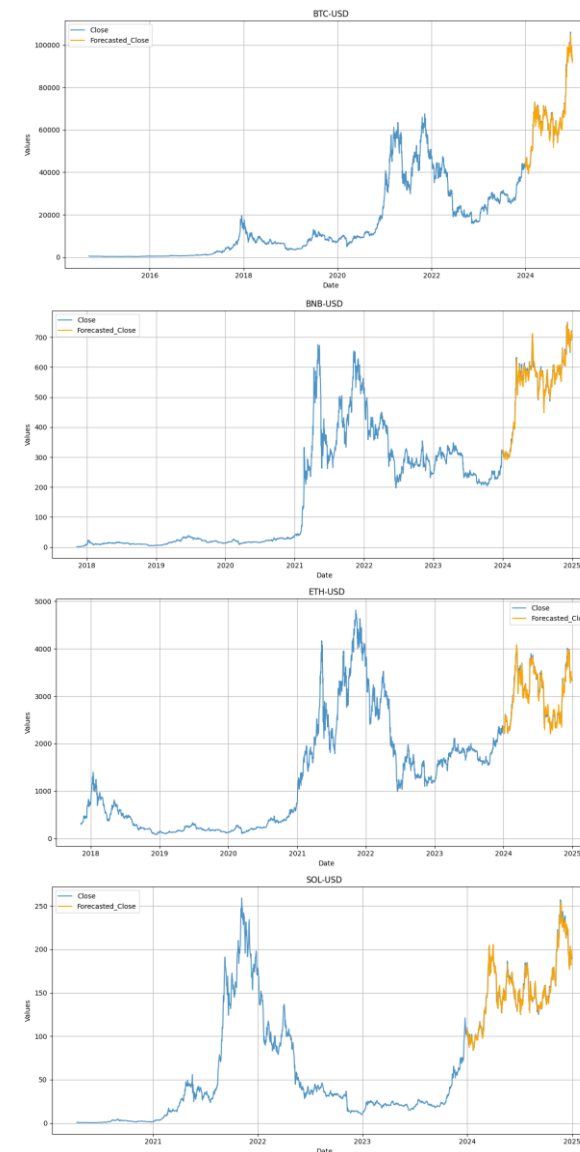
Normalized



Unscaled



Standardized

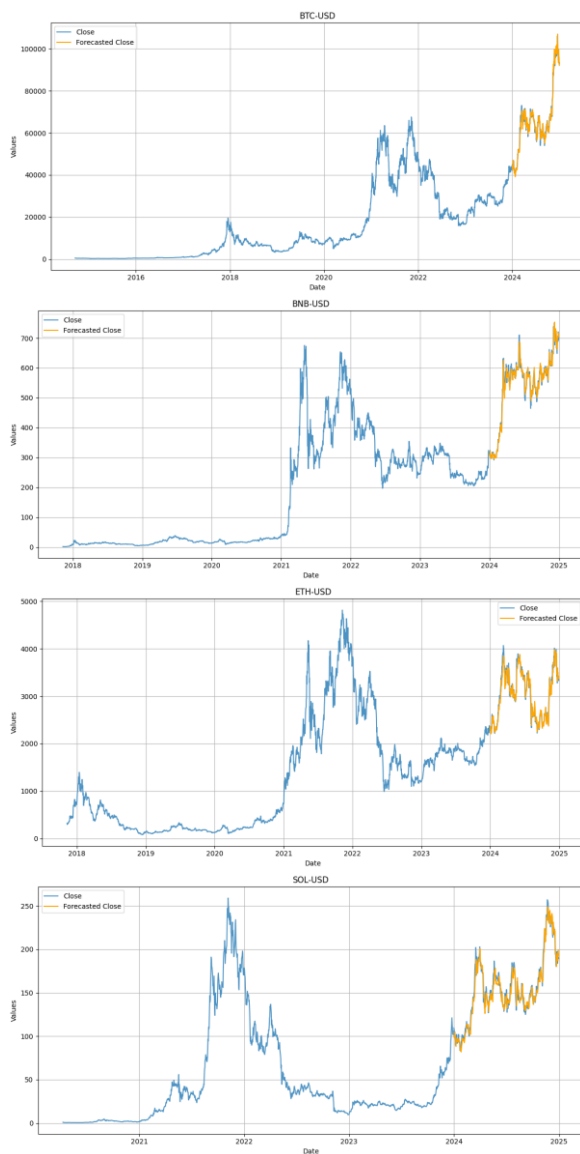


Not a major difference between scaled and unscaled inputs. Standardization seems to be better than Normalization. Scaling is Crucial for finetuning

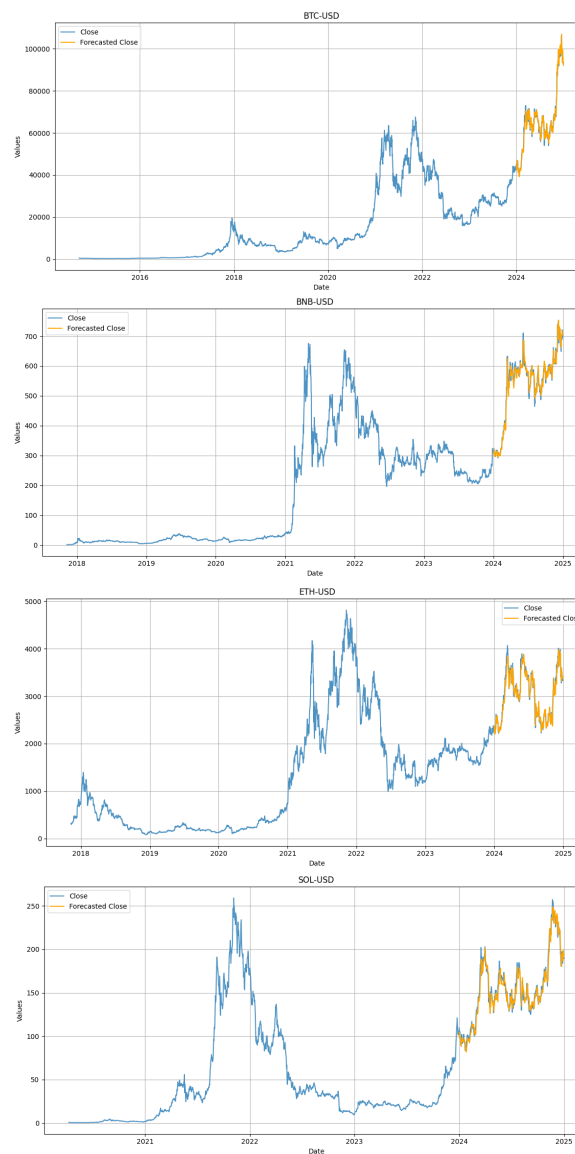
A Note on scaled vs unscaled inputs - IV

Model: **Moirai** | Horizon: 1 | Context Length: 512

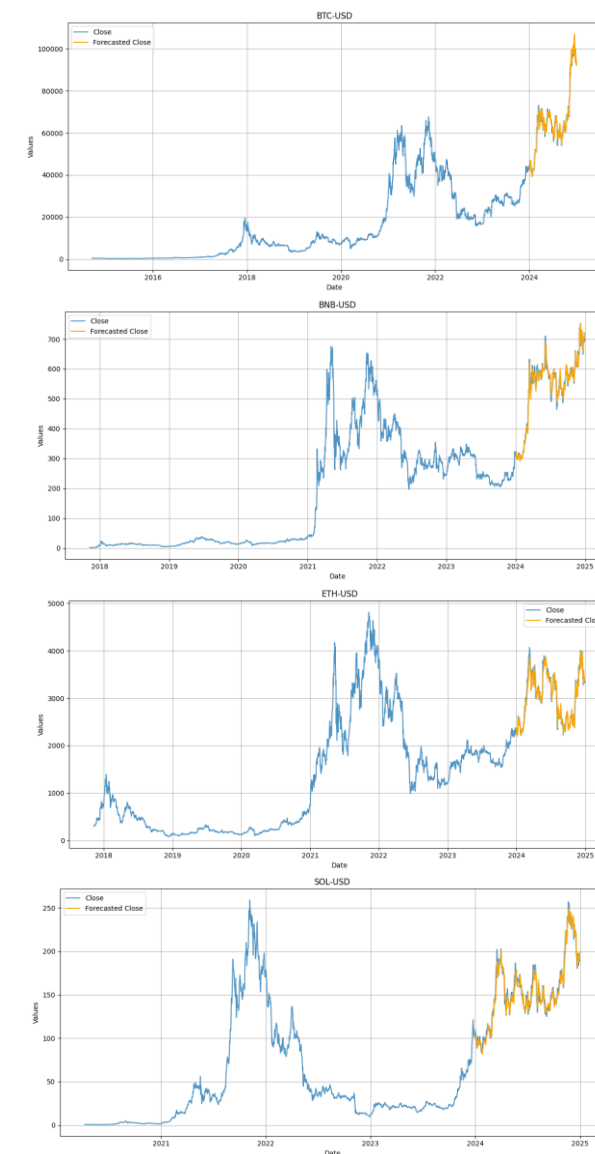
Normalized



Unscaled



Standardized



Not a major difference between scaled and unscaled inputs. Standardization seems to be better than Normalization. Scaling is Crucial for finetuning

A Note on scaled vs unscaled inputs - V

	TimesFM								
	Unscaled			Standardization			Normalization		
Horizon	30	7	1	30	7	1	30	7	1
Metrics									
MAPE	8.6567	5.009	2.5364	8.6569	5.009	2.5364	8.6568	5.009	2.5364
ME	1181.6526	75.7273	-305.0041	1181.6968	75.621	-305.0041	1181.7174	75.6686	-304.9632
MAE	5763.2598	3295.2364	1688.0507	5763.3913	3295.1856	1688.0507	5763.3237	3295.2233	1688.0737
MPE	0.9924	-0.0937	-0.6223	0.9924	-0.0939	-0.6223	0.9924	-0.0938	-0.6222
RMSE	7723.1694	4176.6292	2211.5484	7723.2888	4176.5225	2211.5484	7723.2328	4176.5838	2211.6001
R	0.8565	0.9589	0.9889	0.8565	0.9589	0.9889	0.8565	0.9589	0.9889
Scalar Product	12411252.19	175626978.5	-22417878.21	12408091.24	175652370.7	-22417878.21	12408412.01	175646733.9	-22504554.48
Return Score	0.5096	0.4986	0.4877	0.5096	0.4986	0.4877	0.5096	0.4986	0.4877
Long Return	0.6211	0.5421	0.5158	0.6211	0.5421	0.5158	0.6211	0.5421	0.5158
Short Return	0.3886	0.4514	0.4571	0.3886	0.4514	0.4571	0.3886	0.4514	0.4571
Mean Directional Accuracy	0.5096	0.4986	0.4877	0.5096	0.4986	0.4877	0.5096	0.4986	0.4877
Mean Directional Accuracy Positive	0.6211	0.5421	0.5158	0.6211	0.5421	0.5158	0.6211	0.5421	0.5158
Mean Directional Accuracy Negative	0.3886	0.4514	0.4571	0.3886	0.4514	0.4571	0.3886	0.4514	0.4571

BTC

Note: The performance metrics of scaled series were calculated after inverse transformation

	Chronos								
	Unscaled			Standardization			Normalization		
Horizon	30	7	1	30	7	1	30	7	1
Metrics									
MAPE	7.4117	4.5537	2.0218	8.1217	4.5637	2.0547	7.2328	4.4848	2.0611
ME	775.8249	128.427	41.3087	2163.1399	591.7592	65.5236	995.8332	378.0743	26.5184
MAE	4963.0446	2982.3971	1335.1071	5512.2141	3007.3156	1357.7714	4855.1797	2946.7852	1367.0721
MPE	0.522	0.1772	0.0508	2.6311	0.8454	0.0912	0.9346	0.5031	0.0309
RMSE	7184.7332	3848.7169	1859.8386	8060.0669	3938.3429	1893.8474	7074.7344	3864.9201	1907.5412
R	0.8758	0.9669	0.992	0.8524	0.9654	0.9917	0.8807	0.9662	0.9916
Scalar Product	31949163.89	168698375.7	-22149873.06	-22209285.75	186216629.4	-33636937.53	25664371.73	180262251.5	-76202222.64
Return Score	0.4822	0.4329	0.4932	0.4548	0.4466	0.4849	0.4849	0.411	0.4521
Long Return	0.5474	0.5105	0.5211	0.4474	0.4368	0.5	0.5474	0.4632	0.4579
Short Return	0.4114	0.3486	0.4629	0.4629	0.4571	0.4686	0.4171	0.3543	0.4457
Mean Directional Accuracy	0.4822	0.4329	0.4932	0.4548	0.4466	0.4849	0.4849	0.411	0.4521
Mean Directional Accuracy Positive	0.5474	0.5105	0.5211	0.4474	0.4368	0.5	0.5474	0.4632	0.4579
Mean Directional Accuracy Negative	0.4114	0.3486	0.4629	0.4629	0.4571	0.4686	0.4171	0.3543	0.4457

A Note on scaled vs unscaled inputs - VI

	TTM								
	Unscaled			Standardization			Normalization		
Horizon	30	7	1	30	7	1	30	7	1
Metrics									
MAPE	7.8205	4.6689	2.1309	7.8205	4.3934	2.131	7.8201	4.3933	2.1309
ME	3152.0168	8.8385	470.8715	3151.9762	1009.5409	470.8579	3151.288	1009.2115	470.6378
MAE	5447.2776	25.7264	1406.7461	5447.2613	2921.4697	1406.7449	5446.992	2921.4241	1406.7211
MPE	4.0519	1.5735	0.6405	4.0519	1.3911	0.6405	4.0508	1.3905	0.6402
RMSE	8017.2147	34.3992	1957.2723	8017.1945	3925.5219	1957.2706	8016.8608	3925.4392	1957.2413
R	0.8648	0.9554	0.9915	0.8649	0.9663	0.9916	0.8649	0.9663	0.9916
Scalar Product	-7264848.111	-10257.1327	-52602085.28	-7264715.848	184265929	-52602900.9	-7261490.687	184259395.8	-52608415.83
Return Score	0.4821	0.5041	0.4684	0.4822	0.5178	0.4685	0.4822	0.5178	0.4685
Long Return	0.2684	0.3979	0.5	0.2684	0.4316	0.5	0.2684	0.4316	0.5
Short Return	0.7142	0.6206	0.4342	0.7143	0.6114	0.4343	0.7143	0.6114	0.4343
Mean Directional Accuracy	0.4821	0.5041	0.4684	0.4822	0.5178	0.4685	0.4822	0.5178	0.4685
Mean Directional Accuracy Positive	0.2684	0.3979	0.5	0.2684	0.4316	0.5	0.2684	0.4316	0.5
Mean Directional Accuracy Negative	0.7142	0.6206	0.4342	0.7143	0.6114	0.4343	0.7143	0.6114	0.4343

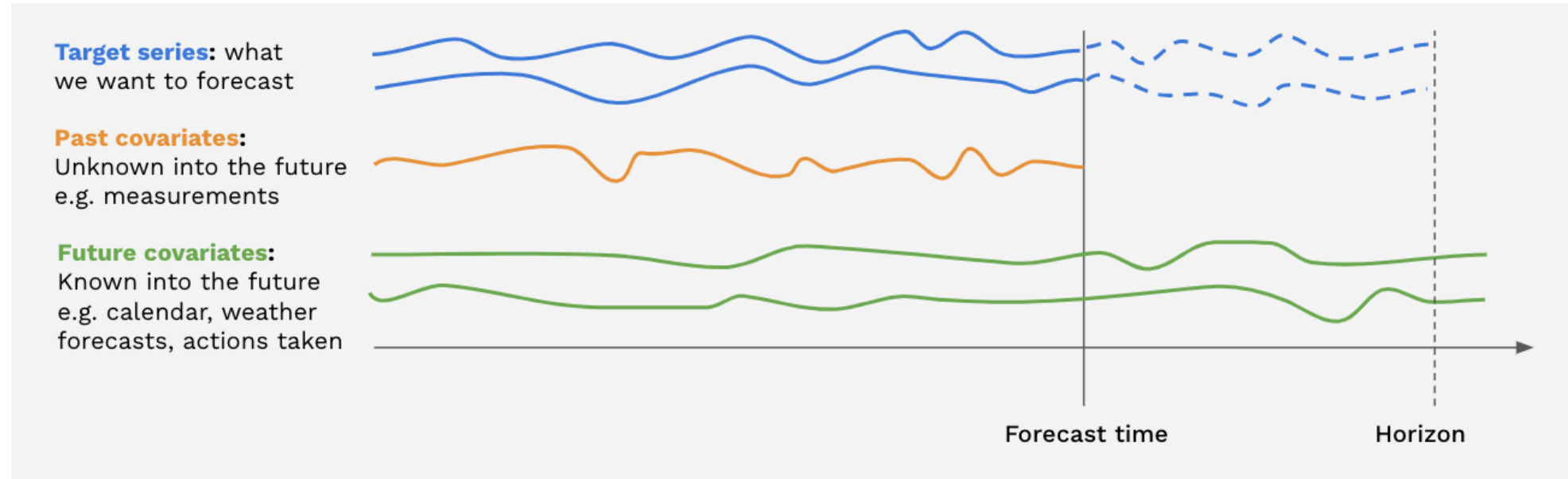
BTC

Note: The performance metrics of scaled series were calculated after inverse transformation

	Moirai								
	Unscaled			Standardization			Normalization		
Horizon	30	7	1	30	7	1	30	7	1
Metrics									
MAPE	7.8042	4.5633	2.3044	7.232	4.5645	2.3056	7.2281	4.5561	2.3049
ME	7.7342	238.6896	239.1631	838.3422	239.1034	239.5822	80035.8846	238.5001	239.1631
MAE	42.8013	2996.6965	1518.4464	4927.727	2997.1788	1518.9213	4925.6125	2992.7438	1518.2981
MPE	1.3692	0.3868	0.378	0.4879	0.3879	0.3791	0.4838	0.3856	0.3767
RMSE	61.8201	3834.2699	2095.1794	7362.6688	3834.4645	2095.5405	7362.1484	3832.112	2094.029
R	0.8639	0.9677	0.9901	0.8673	0.9678	0.9901	0.8673	0.9678	0.9901
Scalar Product	8270.6943	120321755.1	-824247990	41910174.42	120309936	-82438345.87	44450252.7	117909089.4	-81572324.03
Return Score	0.5232	0.4794	0.4712	0.526	0.4795	0.4712	0.5232	0.4739	0.4684
Long Return	0.5497	0.5894	0.4894	0.5737	0.5895	0.4895	0.5736	0.5789	0.4894
Short Return	0.4942	0.36	0.4514	0.4743	0.36	0.4514	0.4685	0.36	0.4457
Mean Directional Accuracy	0.5232	0.4794	0.4712	0.526	0.4795	0.4712	0.5232	0.4739	0.4684
Mean Directional Accuracy Positive	0.5497	0.5894	0.4894	0.5737	0.5895	0.4895	0.5736	0.5789	0.4894
Mean Directional Accuracy Negative	0.4942	0.36	0.4514	0.4743	0.36	0.4514	0.4685	0.36	0.4457

Not a major difference between scaled and unscaled inputs.

Curious Case of Covariates - I



- Given the financial metrics performance, one of the solutions to improve these metrics was to use S&P 500 as a covariate. S&P 500 can serve as a pseudo indicator for the stock market performance
- All the 4 models have amazing performance and benchmarks on zero-shot learning a.k.a univariate time series forecasting
- Though on paper all the 4 models claim to have inclusion of covariates, there are certain crucial limitation when applied in practice
- That is a key reason for not have a reasonable number of study or benchmark on forecasting with covariates for these models

Potential Solution for Covariates Inclusion - I

TimesFM ^[2]

- Support only static covariate and dynamic futuristic covariate

Product: ice cream

Daily_sales: [30, 30, 4, 5, 7, 8, 10]

Category: food

Base_price: 1.99

Weekday: [0, 1, 2, 3, 4, 5, 6, 0, 1, 2, 3, 4, 5, 6]

Has_promotion: [Yes, Yes, No, No, No, Yes, Yes, No, No, No, No, No, No, No]

Daily_temperature: [31.0, 24.3, 19.4, 26.2, 24.6, 30.0, 31.1, 32.4, 30.9, 26.0, 25.0, 27.8, 29.5, 31.2]

- Past Dynamic Covariates is something the batched in context regression cannot address
- Two hacks
 - Shift and repeat these past dynamic covariates to use their delayed version
 - Bootstrap – Run TimesFM/other forecasting model to forecast these past dynamic covariates into the horizon, then call TimesFM again using these forecasts as the future part for these dynamic covariates.

Chronos ^[3]

- Chronos supports predominantly univariate time series forecasting only
- Therefore, it doesn't support covariates by default
- AutoGluon-TimeSeries (AG-TS) enables Chronos to use dynamic covariates & static features by combining it with a tabular regression model
- AutoGluon recommends AWS g5.2xlarge and p3.2xlarge instances that feature NVIDIA A10G and V100 GPUs, with at least 16GiB of GPU memory and 32GiB of main memory for performing covariate analysis

Potential Solution for Covariates Inclusion - II

Tiny Time Mixers^[4]

Moirai^[6]

- As per the literature, TTM and Moirai by default supports multivariate time series forecasting and all type of covariates
- To streamline all the four model, this study has used **sktime** – unified API for ML/AI with time series, for model building, fitting, application, and validation
- Though **sktime** has place holder to support exogenous variables – both past and future covariates, unfortunately, it is not yet integrated with the foundational models like TTM and Moirai
- Potential solution
 - Use Moirai's official code repository - **uni2ts**^[12] to do multivariate analysis.
 - Use IBM's **granite-tsfn**^[13] framework to do multivariate analysis.
 - The initial setup of both these libraries are difficult in windows machine and Google Collab
 - They require specialized OS capabilities to support GPU for faster execution

Takeaway

- Though on literature multivariate time series forecasting using different types of covariates is possible, there a limited practical examples and resources
- Installation of core packages for each model will disrupt the streamline of entire pipeline, hence **sktime** as a framework is used
- Covariate based multivariate forecasting of cryptocurrency prices for each model can be a standalone capstone project
- Covariates need not be restricted to S&P 500 index, a volatile market-based time series like cryptocurrency prices can be use Google Trends, sentiment analysis from social media and News API to have better forecasting

Our Learnings

- Traditional models must be trained on the latest data for each step-up horizon to get reasonable result
- There is no concept of training on latest data for Foundation Models
- A naïve Foundation Model performs reasonably well when compared to benchmark models used on a highly volatile time series like cryptocurrency
- With Domain specific fine tuning Foundation models can achieve better performance
- Context length of the Foundation Model needs past observed data (analogous to training data)
- By standardizing and normalizing, foundational models achieved similar results.
- Evaluation metrics calculation for normalized or standardized data should be done post inverse transformation (on original values)
- Foundation Models can save a lot of time and effort for MVPs
- Optimized and systematic fine tuning can save a lot of cost and time, and increases performance of the model
- Most of the mainstream Foundation models support covariates inclusion on paper but there are limited practical resources and framework support
- Finetuning can enhance the performance and lightweight models like TTM and Moirai promises faster finetuning (within few hours)

Future Work

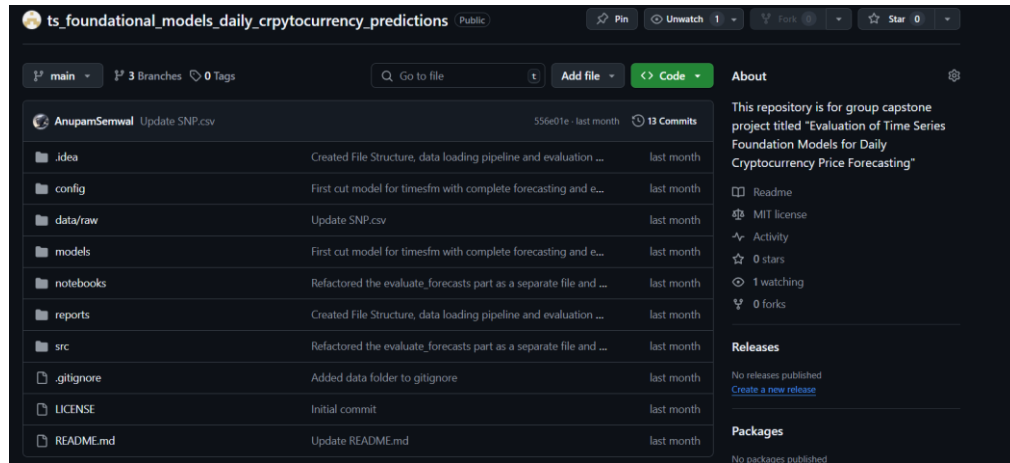
- Inclusion of S&P 500 index as Covariate and track the model performance (financial metrics)
- Explore the multivariate capabilities of the TSFMs and build more practical resources and benchmarks
- Inclusion of market sentiment and news data from social media and news APIs as covariates and track model performance
- Analyze and benchmark top transformer based, non-transformer based and diffusion-based time series foundation models on the benchmarked cryptocurrency data
- Production level forecasting pipeline with Foundation model as the core engine for cryptocurrency forecasting
- Built Benchmark model with seasonality handling and noise removal instead of directly using raw data
- Design a Near-Real time streaming platform for forecasting with configurable models and parameters
- Build a benchmarking open-source ecosystem for TSFMs for easier collaboration



We have already
taken the first step
towards it

A Glimpse of Reusable experimental pipeline

https://github.com/zaskap/ts_foundational_models_daily_cryptoocurrency_predictions



Objective

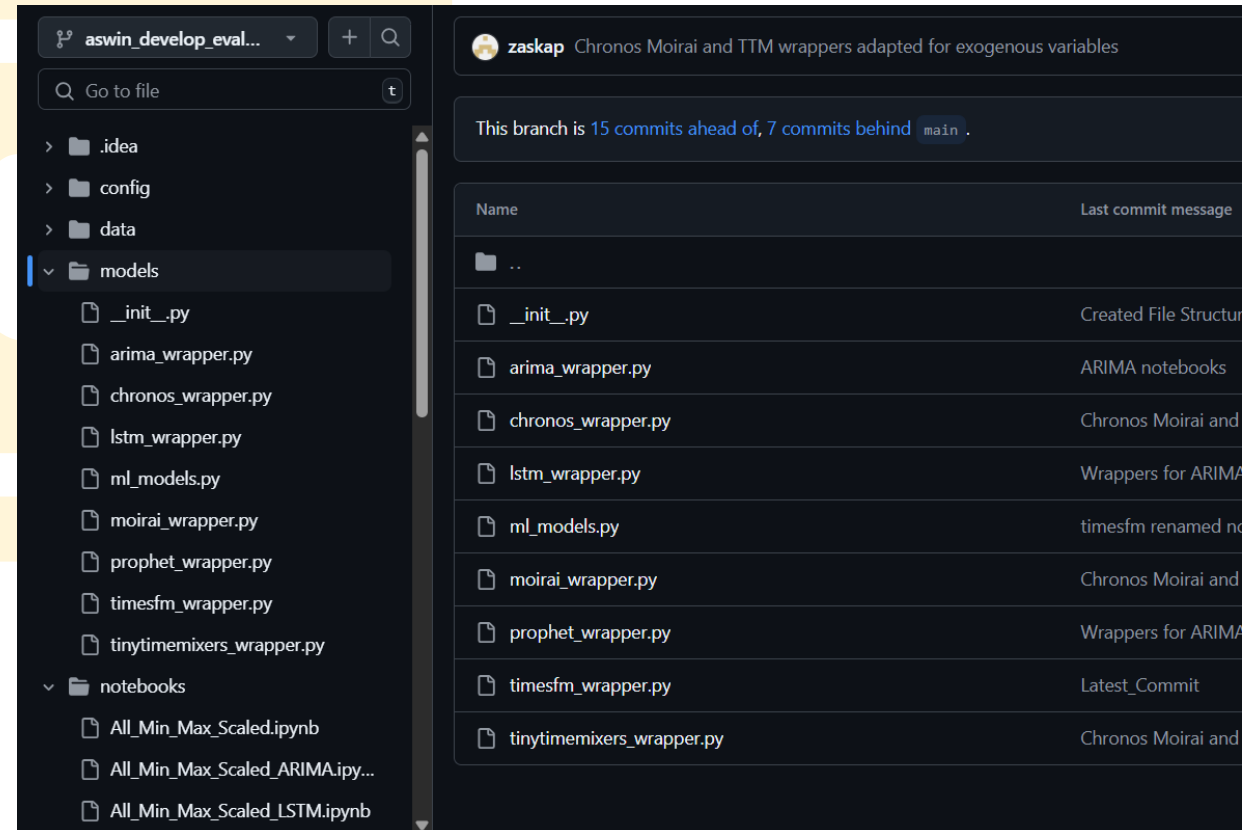
This project focuses on cryptocurrency price forecasting using Time Series Foundation Models. It involves comparing open-source foundational models such as Chronos, TimesFM, Moirai, and Tiny Time Mixers with benchmark models like ARIMA and GRU as suggested in the related works. The study will explore financial and machine learning-based evaluation metrics and analyze the impact of indexation on forecasting performance. The aim of this project is to contribute to the advancement of time series forecasting in the volatile domain of cryptocurrency markets by leveraging innovative foundational models and benchmarking their effectiveness against established forecasting techniques.

Models to be compared

- Chronos [paper](#) | [repo](#)
- TimesFM [paper](#) | [repo](#)
- Moirai [paper](#) | [repo](#)
- Tiny Time Mixers [paper](#) | [repo](#)

Benchmark Models

- ARIMA
- LSTM
- Prophet



References

1. [A Time Series is Worth 64 Words: Long-term Forecasting with Transformers](#)
2. [A decoder-only foundation model for time-series forecasting](#)
3. [Chronos: Learning the Language of Time Series](#)
4. [TTMs: Fast Multi-level Tiny Time Mixers for Improved Zero-shot and Few-shot Forecasting of Multivariate Time Series](#)
5. [Cryptocurrency price forecasting – A comparative analysis of ensemble learning and deep learning methods](#)
6. [A Survey of Time Series Foundation Models: Generalizing Time Series Representation with Large Language Model](#)
7. [Unified Training of Universal Time Series Forecasting Transformers](#)
8. [A decoder-only foundation model for time-series forecasting - A Blog](#)
9. [Forecasting with Chronos – AutoGluon](#)
10. [Chronos: The Rise of Foundation Models for Time Series Forecasting - A Blog](#)
11. [Tiny Time Mixers \(TTM\): A Powerful Zero-Shot Forecasting Model by IBM - A Blog](#)
12. [Salesforce AI Research's uni2ts Framework](#)
13. [IBM's granite-tsfm Framework](#)

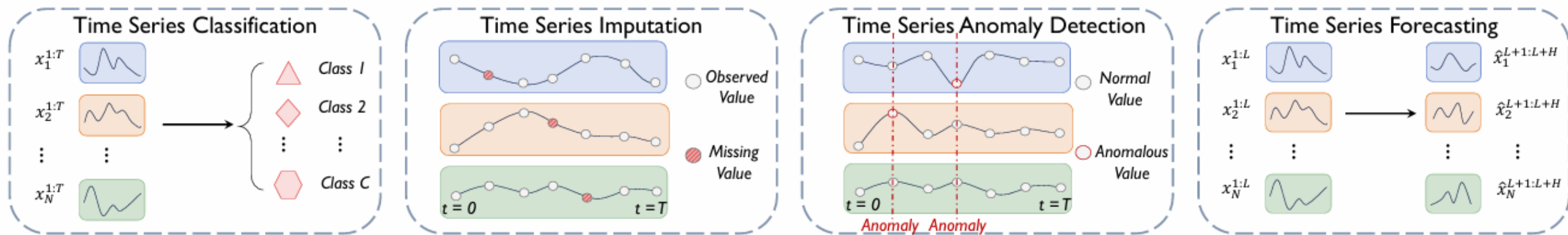


Thank You!
Any Questions

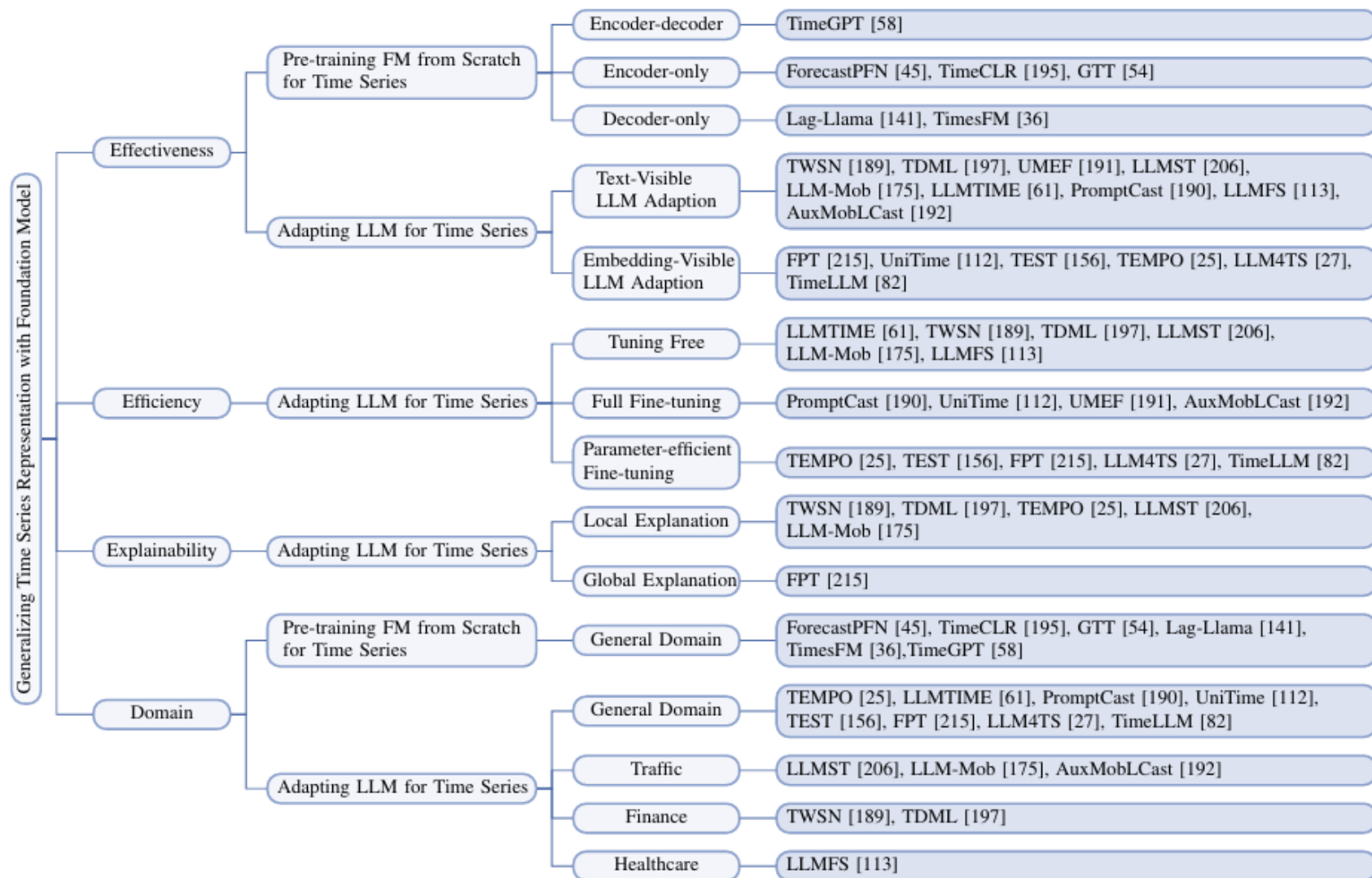


Appendix

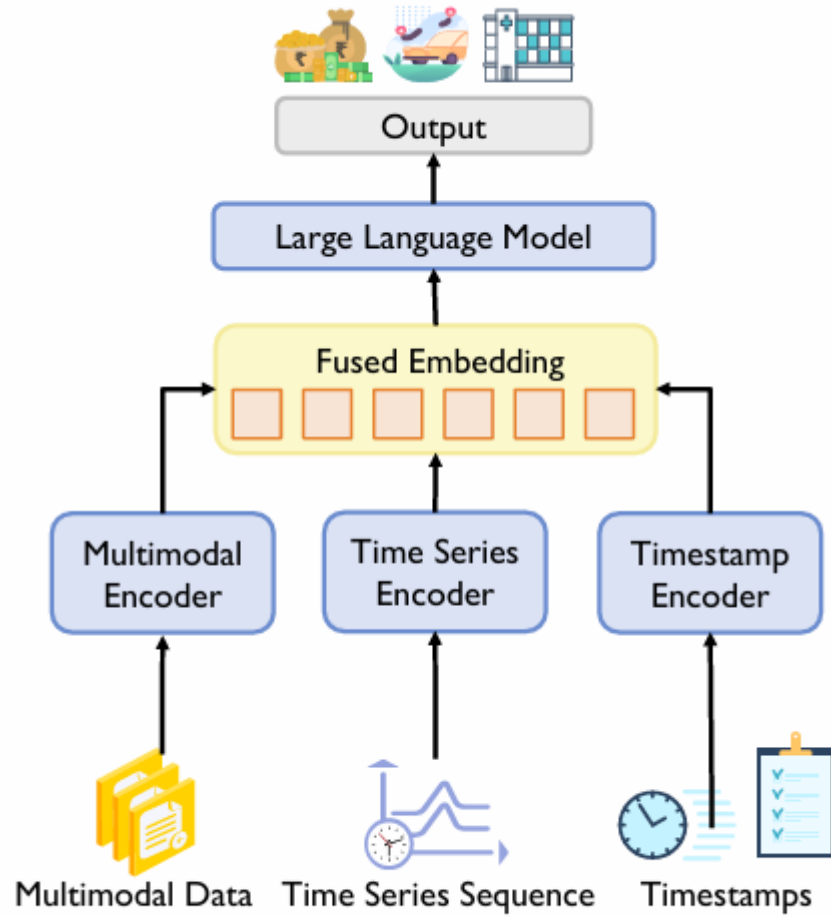
Typical Time Series Tasks^[6]



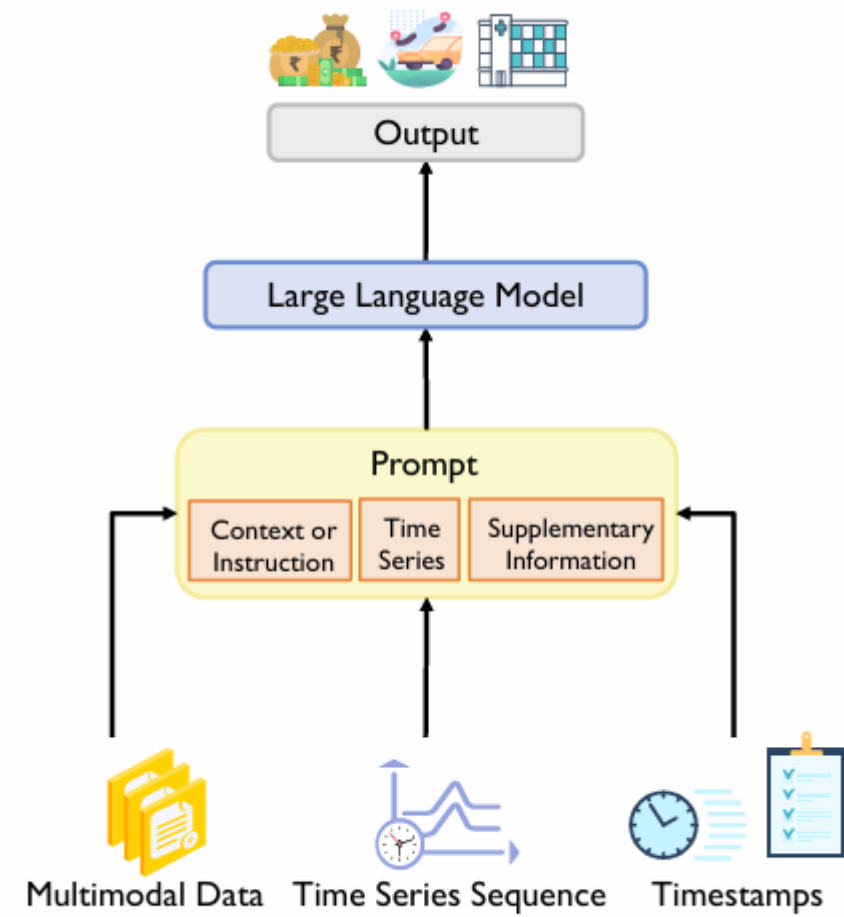
Scope for Foundation Model Research^[6]



LLM adaption paradigm for Time Series tasks ^[6]

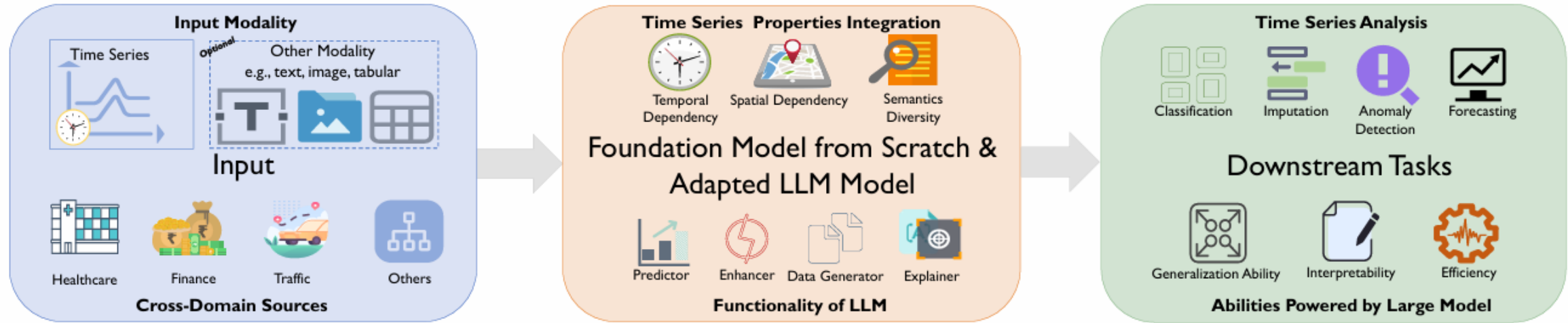


(a) Embedding Visible Large Language Models

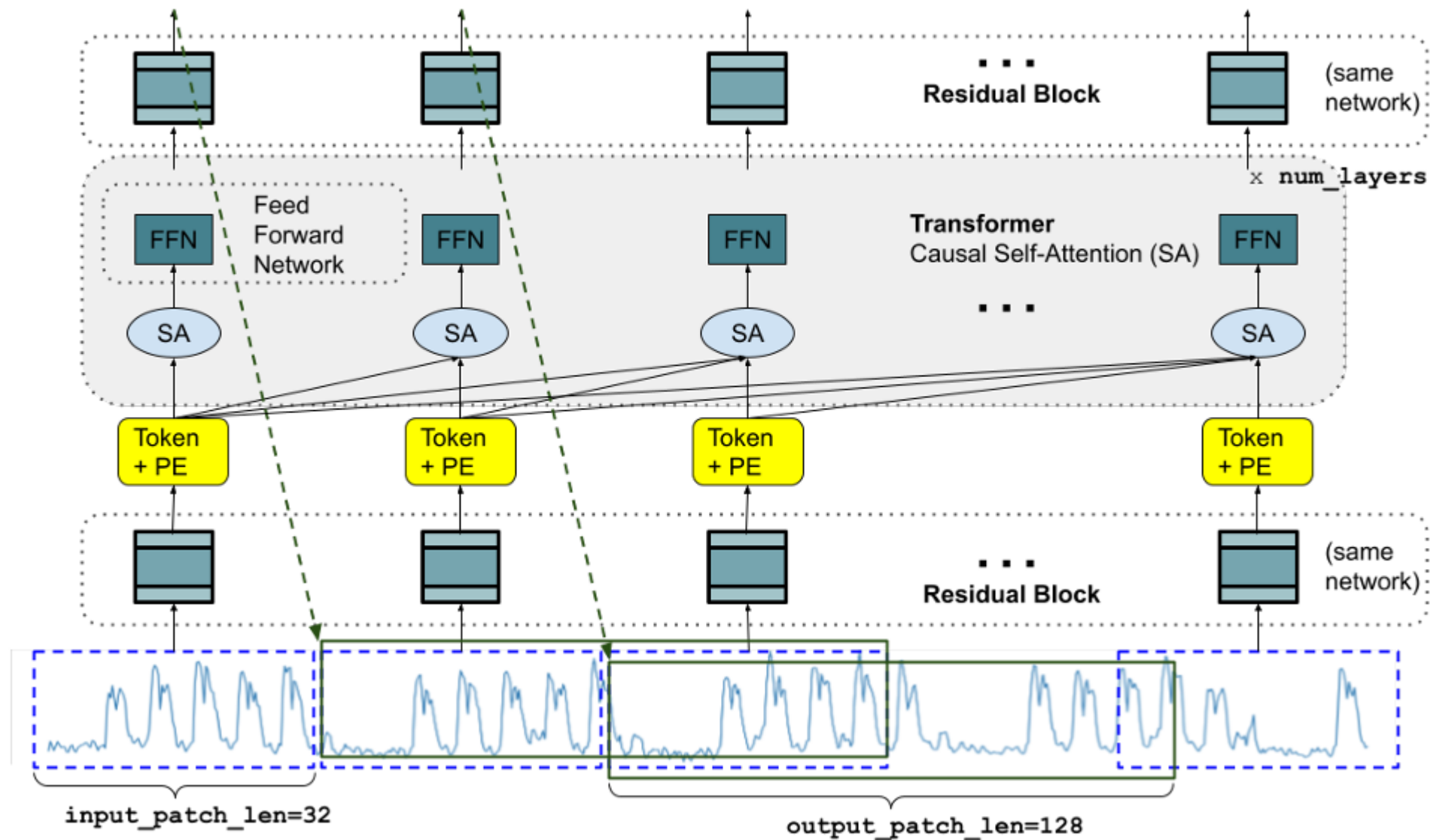


(b) Text Visible Large Language Models

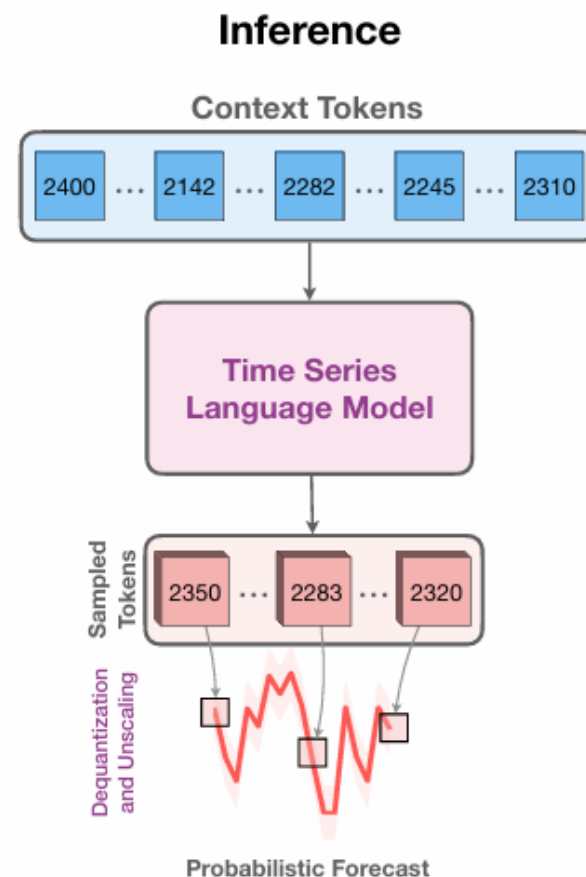
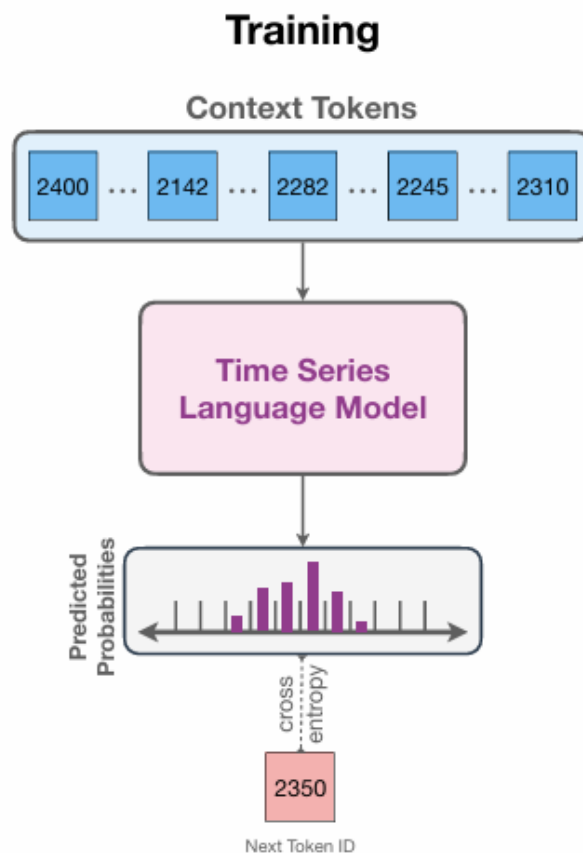
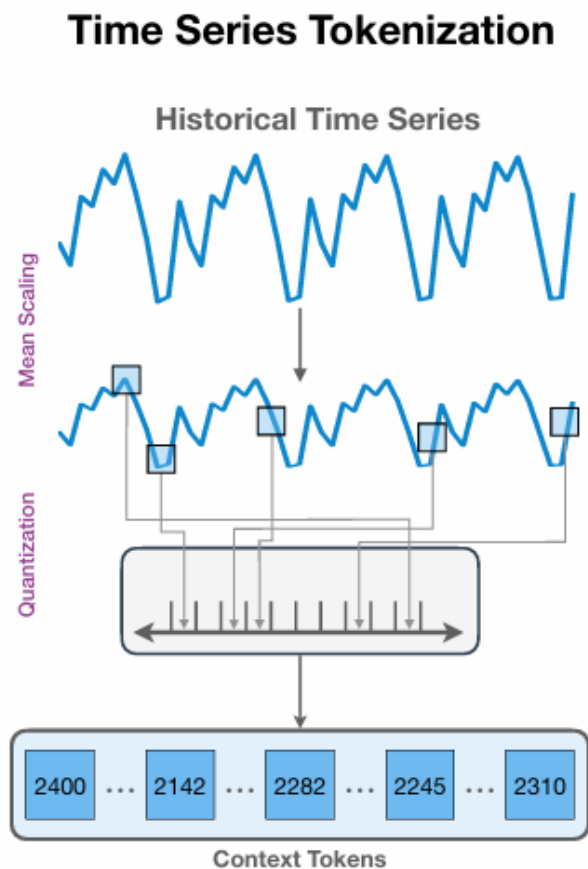
Generalizing Capabilities of Foundation Models^[6]



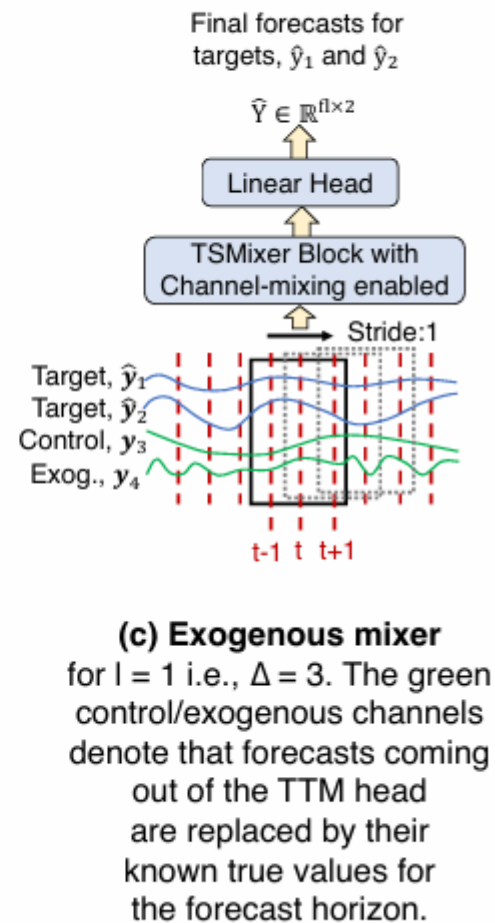
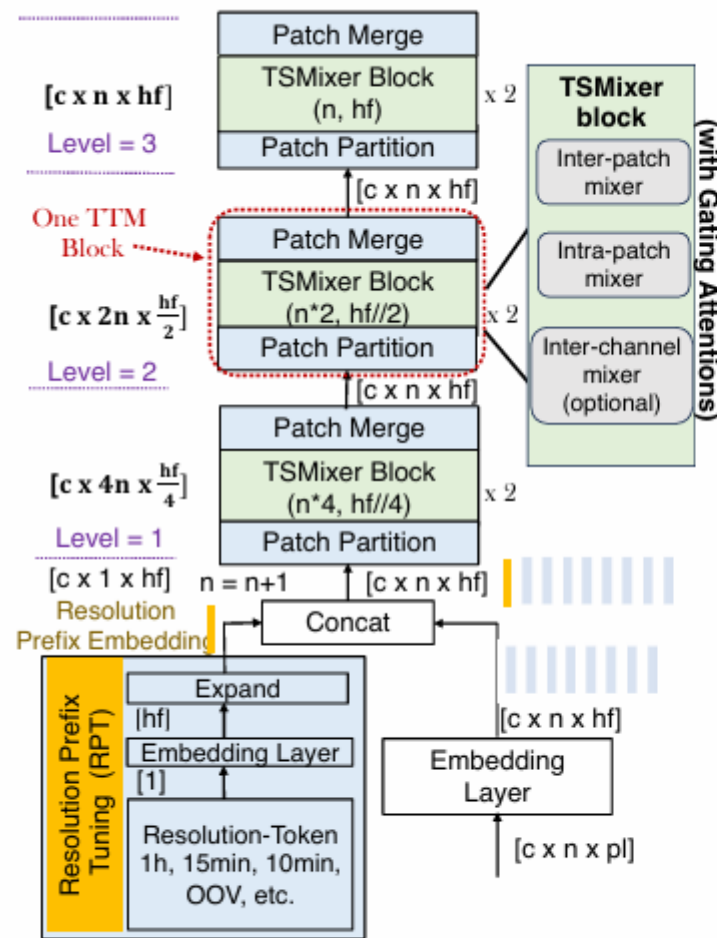
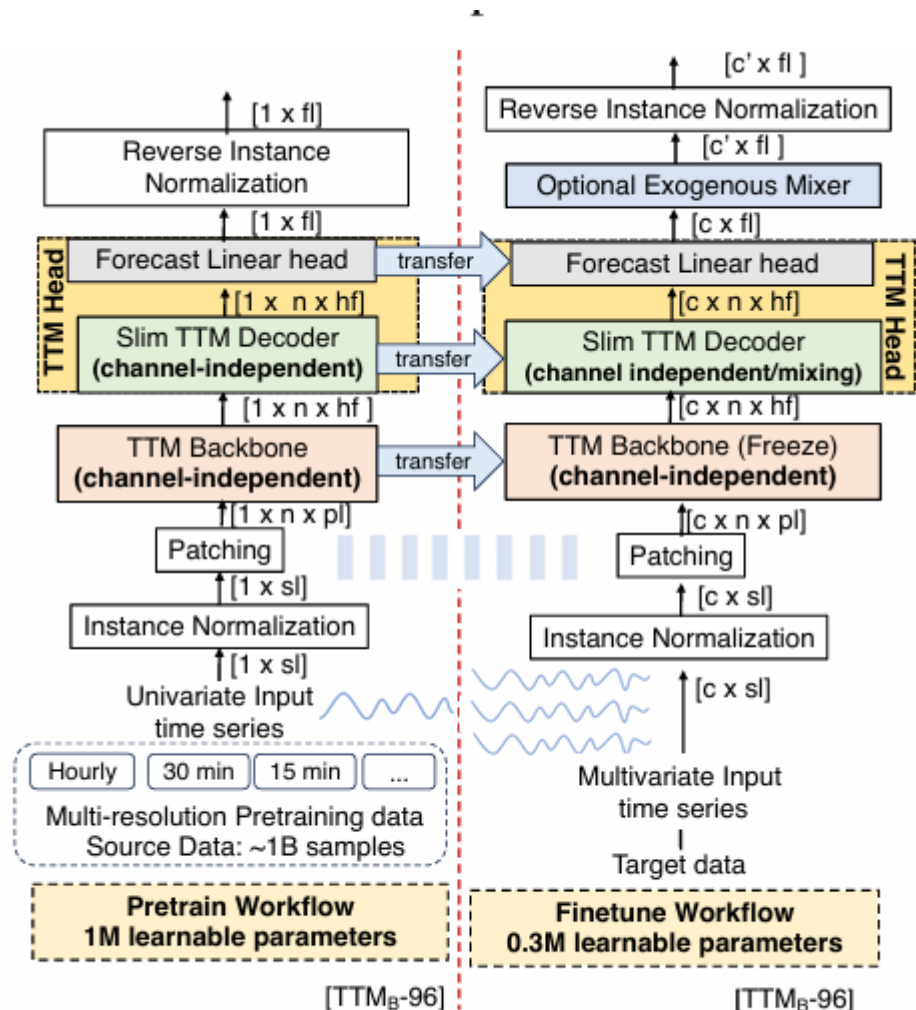
TimesFM Architecture Reference^[2]



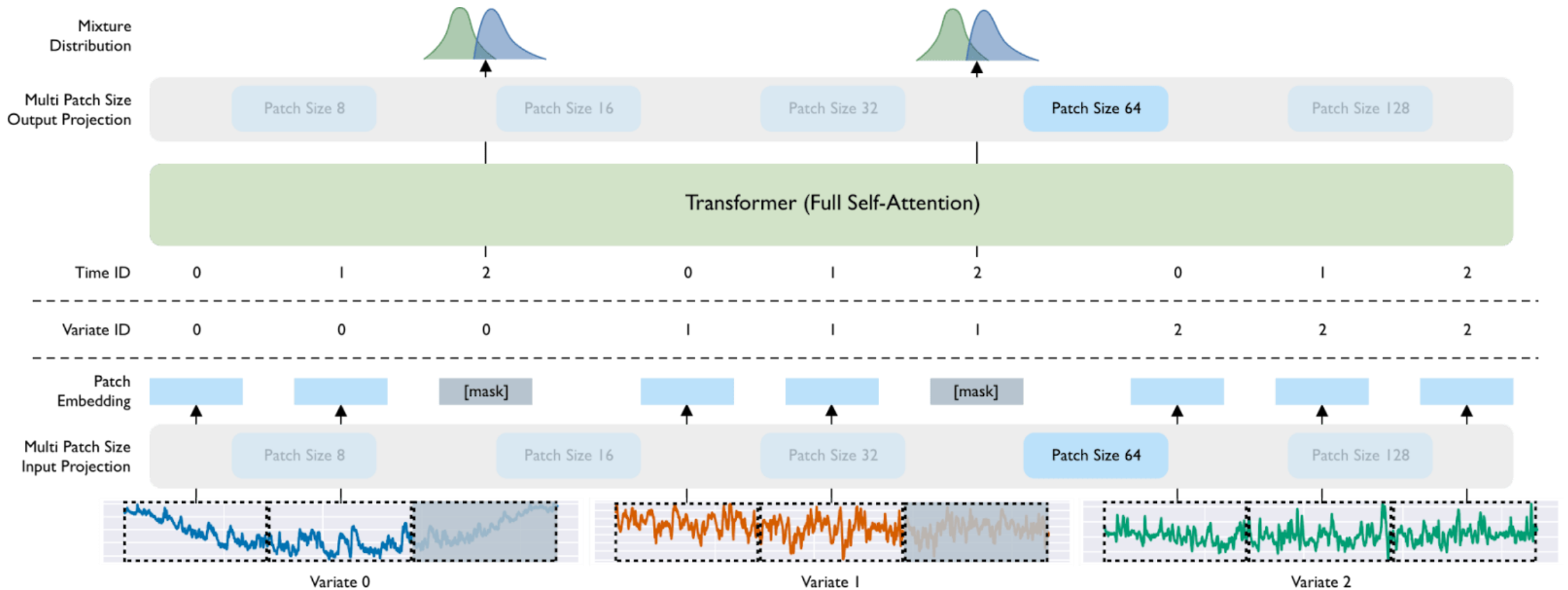
High Level Depiction of CHRONOS^[3]



Tiny Time Mixers Overview^[4]



Moirai Overview^[7]



- The visualization depicts a 3-variate time series, where variates 0 and 1 represent target variables (i.e., those to be forecasted), and variate 2 serves as a dynamic covariate (with known values in the forecast horizon).
- Utilizing a patch size of 64, each variate is patchified into three tokens. These patch embeddings, along with sequence and variate identifiers, are fed into the Transformer. The shaded patches in the visualization denote the forecast horizon to be predicted.
- The corresponding output representations of these patches are then mapped into the parameters of the mixture distribution.